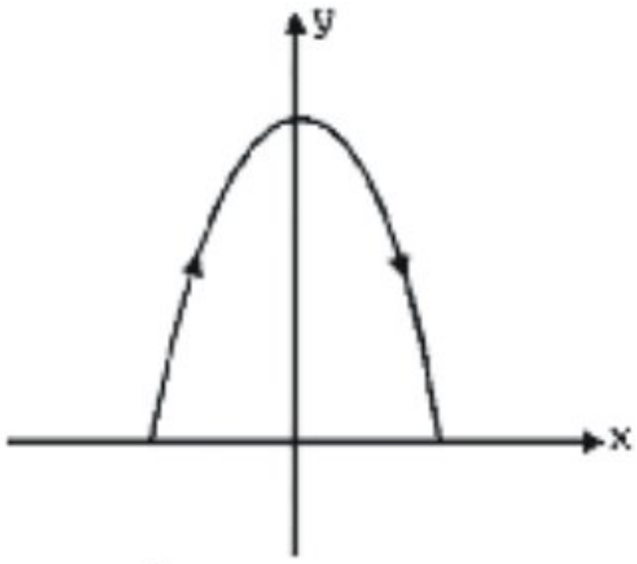
PHYSICS

1)

A wire carrying a current of 4A is bent in the form of a parabola $x^2 + y = 16$ as shown in figure, where x and y are in meter. The wire is placed in a uniform magnetic field $\vec{B} = 5\hat{k}$ tesla. The force acting on the wire is



(1) $80\hat{j}$ N

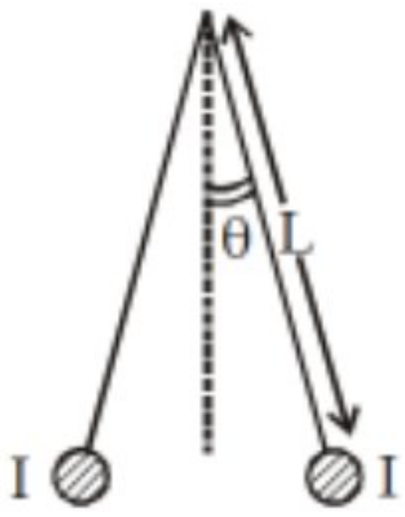
(2) $-80\hat{j}$ N

(3) $-160\hat{j}$ N

(4) $160\hat{j}$ N

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2) Two long current carrying thin wires, both with current I, are held by the insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle 'θ' with the vertical. If wires have mass λ per unit length then the value of I is :- (g = gravitational acceleration)



(1) $2\sqrt{\frac{\pi g L}{\mu_0}\tan\theta}$

(2) $\sqrt{\frac{\pi \lambda g L}{\mu_0}\tan\theta}$

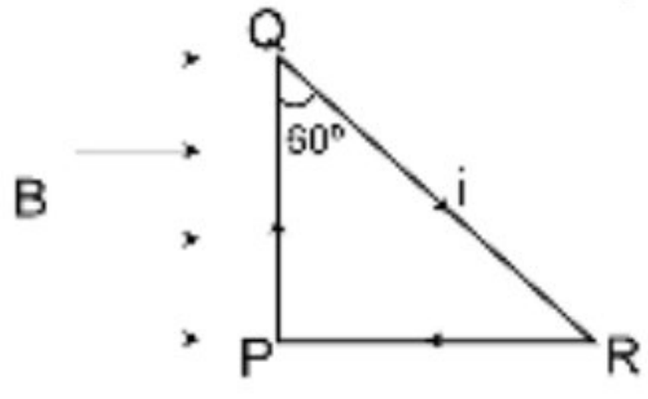
(3) $\sin\theta\sqrt{\frac{\pi \lambda g L}{\mu_0 \cos\theta}}$

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(4) $2\sin\theta\sqrt{\frac{\pi\lambda gL}{\mu_0\cos\theta}}$

3) For the circuit shown in figure, the direction and magnitude of the force on PQR is :



(1) No resultant force act on the loop

(2) ILB out of the page

(3) $\frac{1}{2}$ILB into the page

(4) ILB into the page

4) A long straight wire along the z-axis carries a current I in the negative z-direction. The magnetic vector field $\vec{B}$ at a point having coordinates (x,y) in the z = 0 plane is :

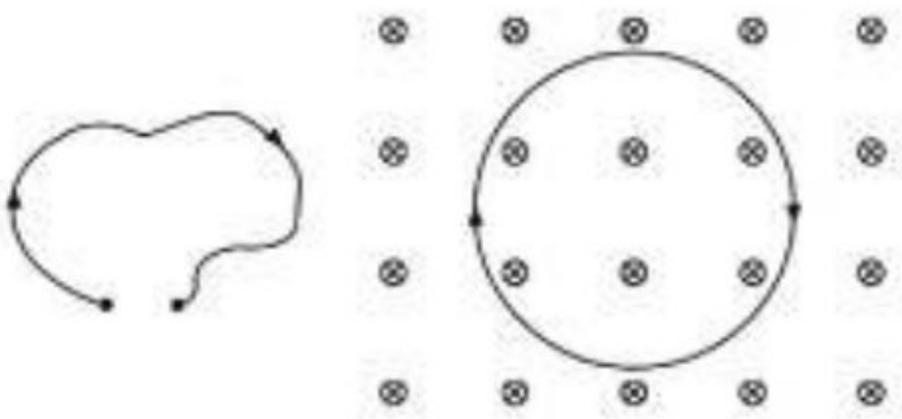
(1) $\frac{\mu_0 I\left(y\hat{i}-x\hat{j}\right)}{2\pi\left(x^2+y^2\right)}$

(2) $\frac{\mu_0 I\left(x\hat{i}+y\hat{j}\right)}{2\pi\left(x^2+y^2\right)}$

(3) $\frac{\mu_0 I\left(x\hat{j}+y\hat{i}\right)}{2\pi\left(x^2+y^2\right)}$

(4) $\frac{\mu_0 I\left(x\hat{i}+y\hat{j}\right)}{2\pi\left(x^2+y^2\right)}$

5) A thin flexible wire of length L is connected to two adjacent fixed points carries a current I in the clockwise direction, as shown in figure.



When system is put in a uniform magnetic field of strength B going into the plane of paper, the wire takes the shape of a circle. The tension in the wire is:

(1) IBL

(2) $\frac{IBL}{\pi}$

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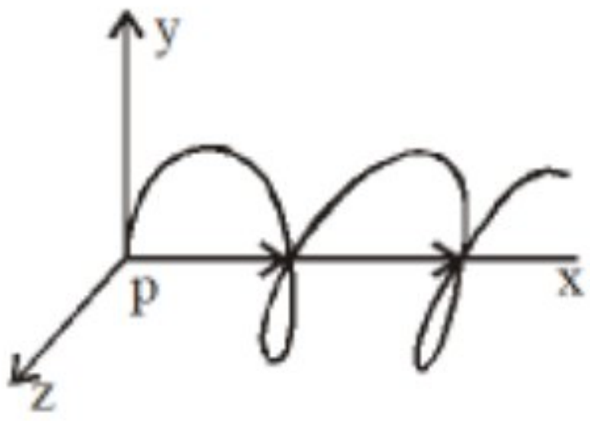
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(3) $\frac{IBL}{2\pi}$

(4) $\frac{IBL}{4\pi}$

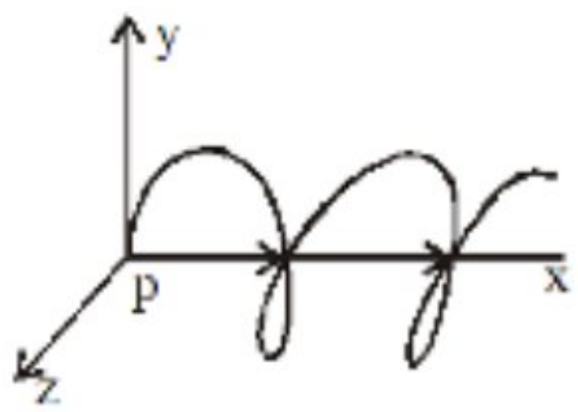
6) An electron is projected from origin with velocity $3.2\hat{i} + \hat{j}$ (m/s). A uniform magnetic field of $2\hat{k}$ tesla is present. The path followed by electron is correctly shown in ($m_e = 9 \times 10^{-31}$ kg)

(1)



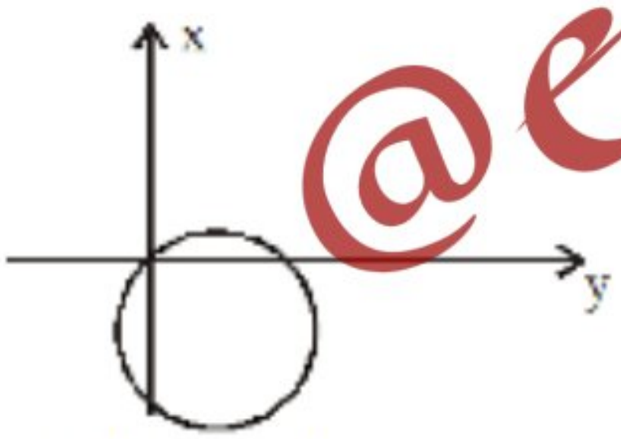
Helix on +ve side of z-axis

(2)



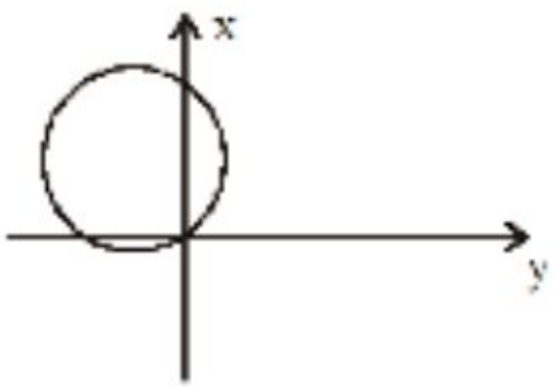
Helix on -ve side of z-axis

(3)



circle in xy plane

(4)



circle in xy plane

7) The magnetic induction at the centre O of the current carrying bent wire shown in the adjoining

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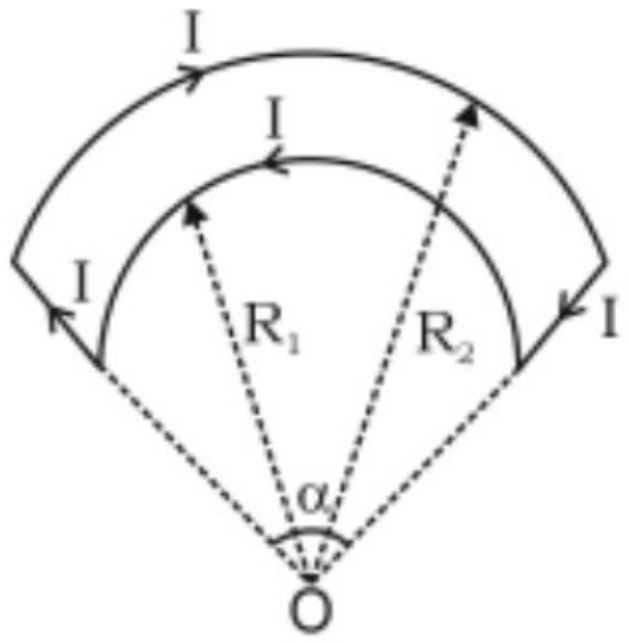


figure is:

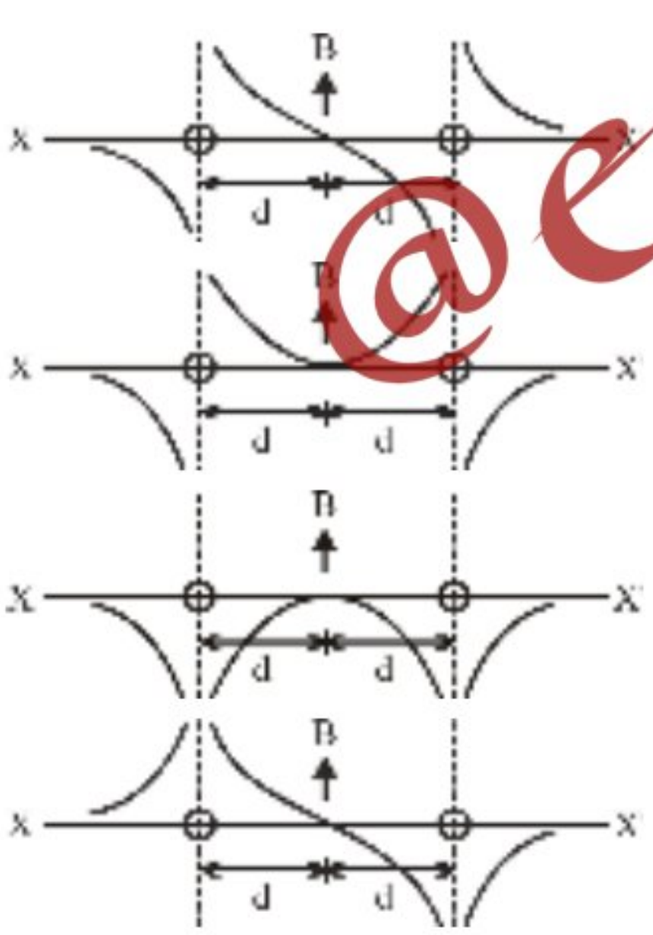
(1) $\frac{\mu_0 I}{4\pi R_1}\alpha$

(2) $\frac{\mu_0 I}{4\pi R_2}\alpha$

(3) $\frac{\mu_0 I\alpha}{4\pi}\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

(4) $\frac{\mu_0 I\alpha}{4\pi}\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$

8) Two long parallel wires are at a distance 2d apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of magnetic field B along the line XX' is given by :

(1) 

(2)

(3)

(4)

9) A thin circular disk of radius R is uniformly charged with density σ > 0 per unit area. The disk rotates about its axis with a uniform angular speed ω. The magnetic moment of the disk is :-

(1) $2\pi R^4\sigma\omega$

(2) $\pi R^4\sigma\omega$

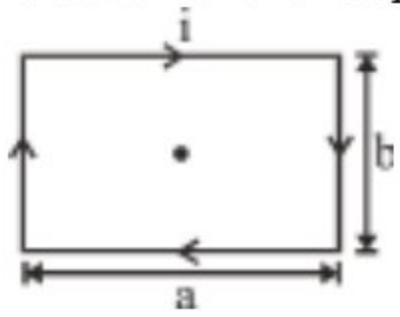
(3) $\frac{\pi R^4}{2}\sigma\omega$

(4) $\frac{\pi R^4}{4}\sigma\omega$

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10) Adjoining figure shows a rectangular loop of conductor carrying a current i. The length and breadth of the loop are respectively a and b. The magnetic field at the centre of loop is :-



(1) $\frac{\mu_0 i(a+b)}{2\pi\sqrt{a^2+b^2}}$

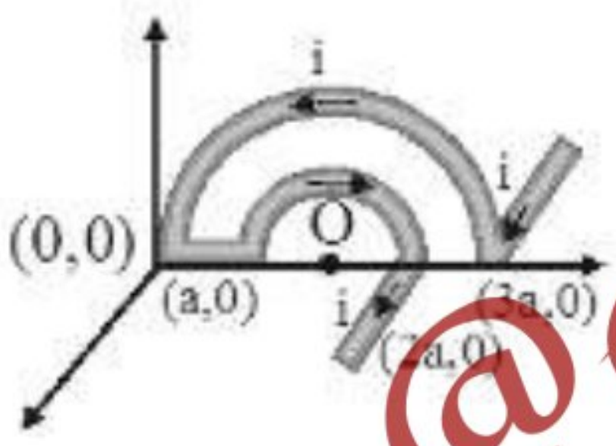
(2) $\frac{\mu_0 iab}{2\pi\sqrt{a^2+b^2}}$

(3) $\frac{\mu_0 i(a+b)}{\pi\sqrt{a^2+b^2}}$

(4) $\frac{2\mu_0 i\sqrt{a^2+b^2}}{\pi ab}$

11)

In the given figure net magnetic field at O will be:-



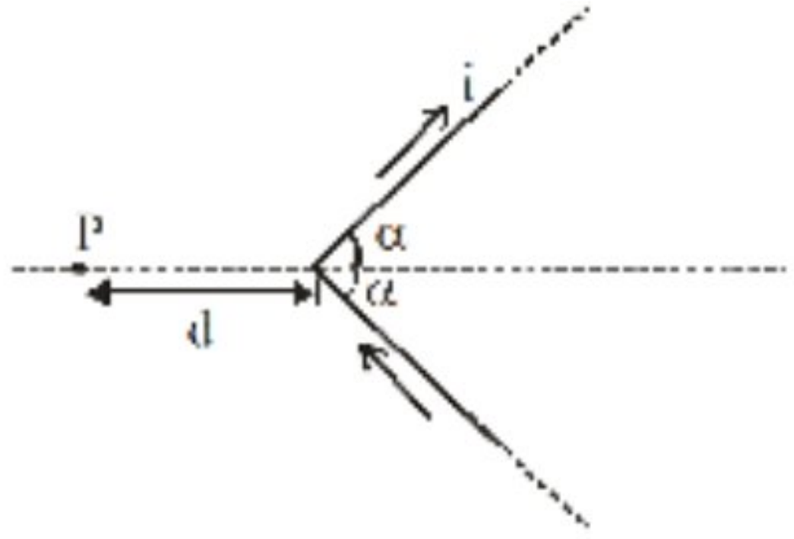
(1) $\frac{2\mu_0 i}{3\pi a}\sqrt{4-\pi^2}$

(2) $\frac{\mu_0 i}{3\pi a}\sqrt{4+\pi^2}$

(3) $\frac{2\mu_0 i}{3\pi a^2}\sqrt{4+\pi^2}$

(4) $\frac{4\mu_0 i}{3\pi a}\sqrt{4-\pi^2}$

12) If the magnetic field at 'P' in the given figure can be writtten as K $\tan\left(\frac{\alpha}{2}\right)$ then K is ;



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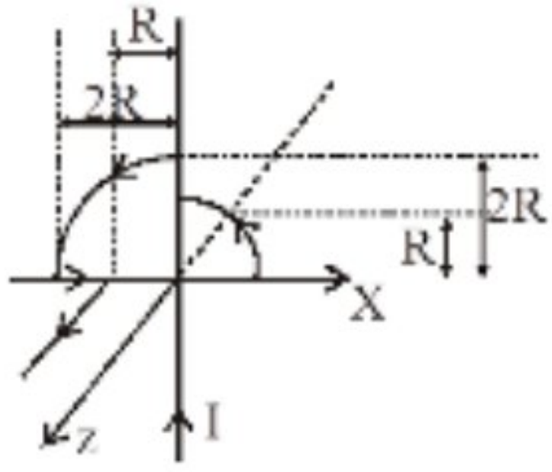
(1) $\frac{\mu_0 i}{4\pi d}$

(2) $\frac{\mu_0 i}{2\pi d}$

(3) $\frac{\mu_0 i}{\pi d}$

(4) $\frac{2\mu_0 i}{\pi d}$

13) Find the magnetic induction at the origin in the figure shown.



(1) $\frac{\mu_0 I}{2R}\left(\frac{3}{4}\hat{k}+\frac{1}{\pi}\hat{j}\right)$

(2) $\frac{\mu_0 I}{4R}\left(\frac{3}{4}\hat{k}-\frac{1}{\pi}\hat{j}\right)$

(3) $\frac{\mu_0 I}{4R}\left(\frac{3}{2}\hat{k}-\frac{1}{\pi}\hat{j}\right)$

(4) $\frac{\mu_0 I}{4R}\left(\frac{3}{4}\hat{k}+\frac{1}{\pi}\hat{j}\right)$

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14) Two identical charged particles enter a uniform magnetic field with same speed but at angles 30° and 60° with field. Let a, b and c be the ratio of their time periods, radii and pitches of the helical paths than

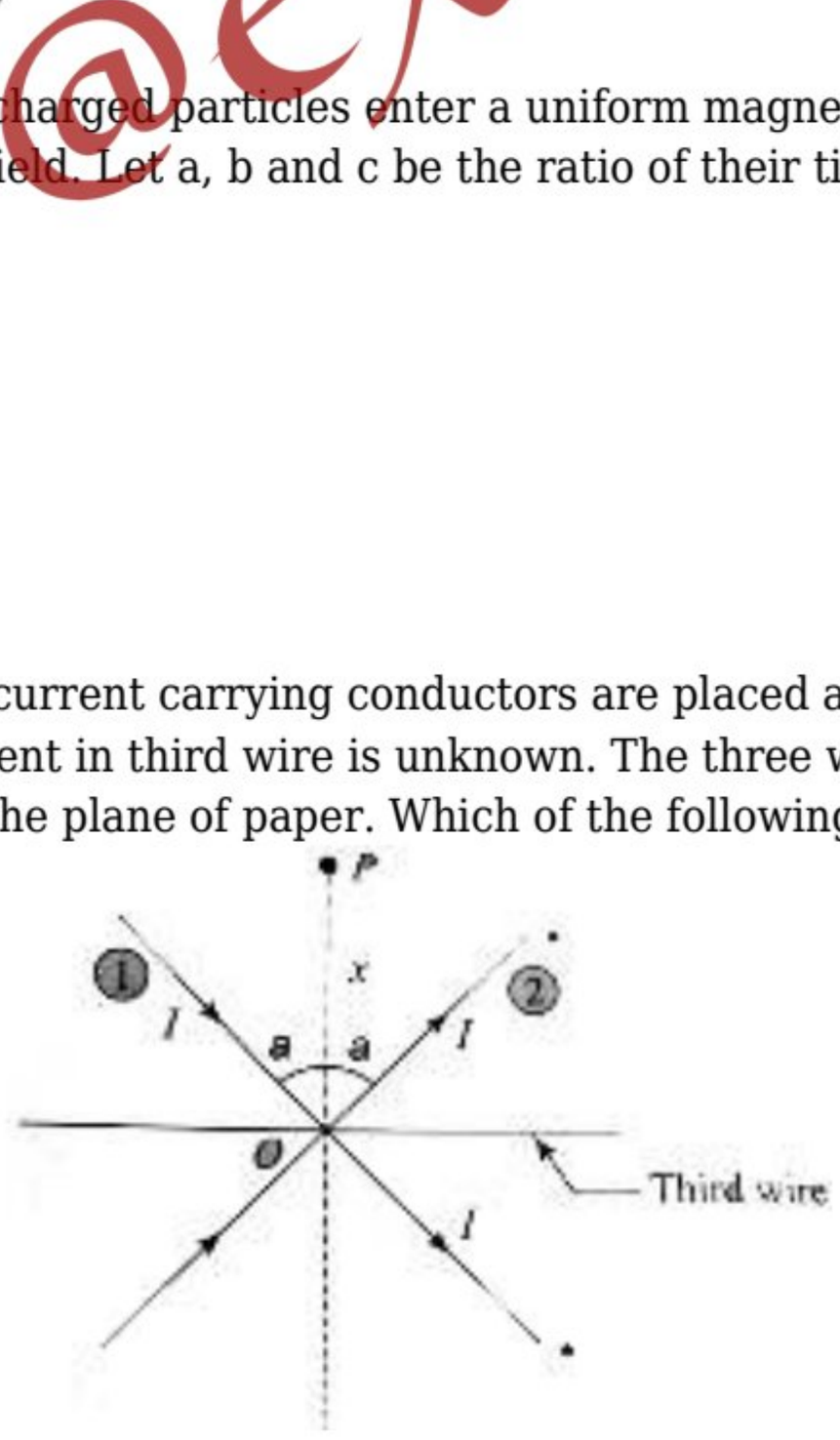
(1) abc = 1

(2) abc > 1

(3) abc < 1

(4) None of these

15) Three infinite current carrying conductors are placed as shown in figure. Two wires carry same current while current in third wire is unknown. The three wires do not intersect with each other and all of them are in the plane of paper. Which of the following is correct about a point 'P' which is also in the same plane:



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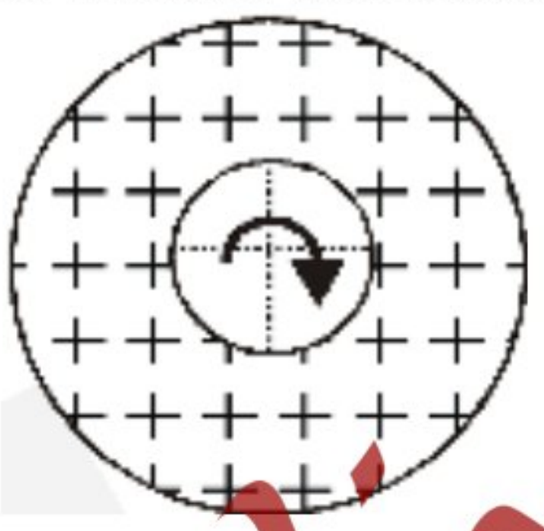
(1) Magnetic field intensity at P is zero for all values of x.

(2) if the current in the third wire is $\frac{2I}{\sin\alpha}$ (left to right) then magnetic field will be zero at P for all values of x.

(3) If the current in the third wire is $\frac{2I}{\sin\alpha}$ (right to left) then magnetic field will be zero at P for all values of x.

(4) None of these.

16) The diagram shows a uniformly charged thin circular disc. The charge per unit area is σ. The inner radius is 'a' and the outer radius is 'b'. The disc is rotated with constant angular velocity ω.



Find out the magnetic field strength at the center.

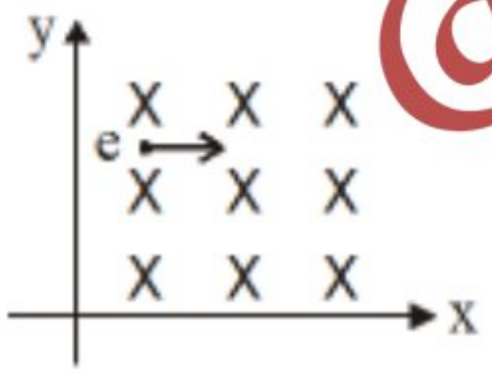
(1) $\mu_0\sigma\omega(b - a)/2$

(2) $\mu_0\sigma\omega\pi(b^2 - a^2)/2$

(3) $\mu_0\sigma\omega(b - a)/2\pi$

(4) None

17) In the given figure the electron enters into the magnetic field. It deflects in direction :



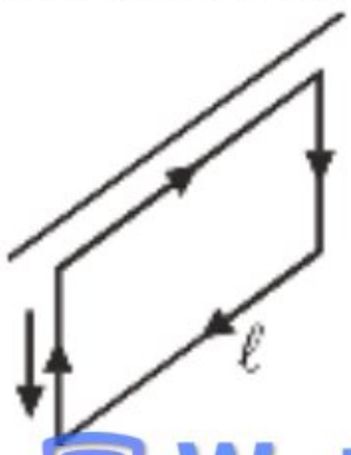
(1) positive X-direction

(2) negative X-direction

(3) positive Y-direction

(4) negative Y-direction

18) A square cardboard of side □ and mass m is suspended from a horizontal axis XY as shown in figure. A single wire is wound along the periphery of board and carrying a clockwise current I. At t=0, a vertical downward magnetic field of induction B is switched on. Find the minimum value of B so that the board will be able to rotate up to horizontal level.



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(1) $\frac{mg}{2I\ell}$

(2) $\frac{2mg}{I\ell}$

(3) $\frac{mg\ell}{2I}$

(4) $\frac{mg}{3I\ell}$

19) A solenoid of length 0.4 m and having 500 turns of wire carries a current of 3 amp. A thin coil having 10 turns of wire and of radius 0.01 m carries a current of 0.4 amp. Calculate the torque required to hold the coil in the middle of the solenoid with its axis perpendicular to the axis of solenoid :-

(1) 8×10^{-10} N–M

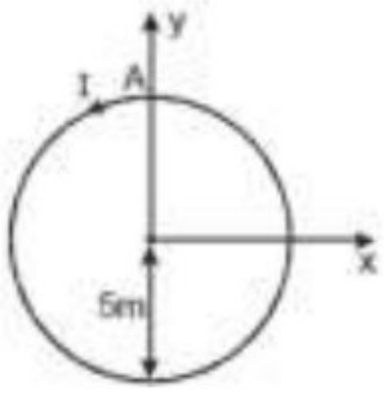
(2) 6×10^{-10} N–M

(3) 6×10^{-6} N–M

(4) 8×10^{-12} N–M

20)

A ring of radius 5m is lying in the x-y plane and is carrying current of 1 A in anti-clock wise sense. If a uniform magnetic field $\vec{B} = 3\hat{i} + 4\hat{j}$ is switched on, then the co-ordinates of point about which the loop will lift up is:



(1) (3, 4)

(2) (4, 3)

(3) (3, 0)

(4) (–3, –4)

21) A 5 cm × 12 cm coil with number of turns 600 is placed in a magnetic field of strength 0.10 Tesla. The maximum magnetic torque acting on it when a current of 10^{-5} A is flowing through it will be

(1) 3.6×10^{-6} N-m

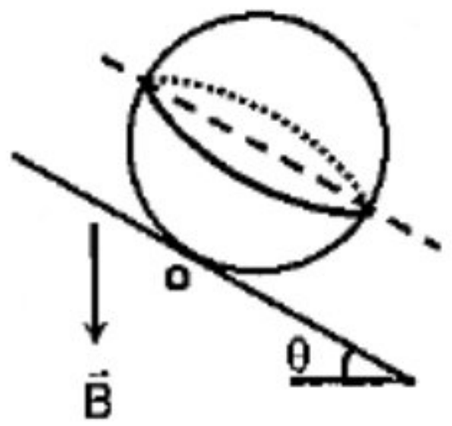
(2) 3.6×10^{-6} dyne-cm

(3) 3.6×10^{6} N-m

(4) 3.6×10^{6} dyne-m

22) In the figure shown, a coil of single tum is wound on a sphere of radius r and mass m. The plane of the coil is parallel to the inclined plane and lies in the equatorial plane of the sphere. It the sphere

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is in rotational equilibrium, the value of B is (current in the coil is i)

(1) $\frac{mg}{\pi ir}$

(2) $\frac{mg\sin\theta}{\pi i}$

(3) $\frac{mgr\sin\theta}{\pi i}$

(4) None of these

23) Two bar magnets having same geometry with magnetic moments M and 2M, are firstly placed in such a way that their similar poles are on same side. Then, its time period of oscillation is T_1. Now, the polarity of the one of the magnet is reversed. Then, time period of oscillation is T_2, then

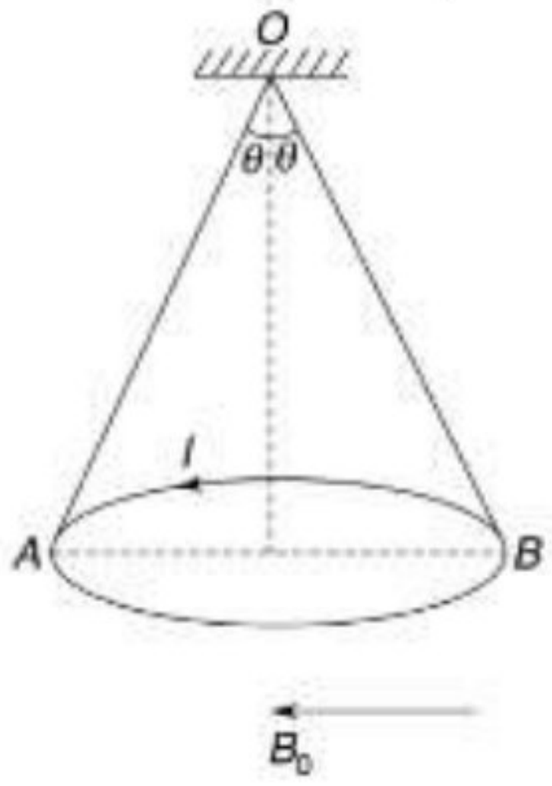
(1) $T_1 < T_2$

(2) $T_1 = T_2$

(3) $T_1 > T_2$

(4) $T_2 = \infty$

24) A uniform ring of mass M and radius R carries a current I (see figure). The ring is suspended using two identical strings OA and OB. There exists a uniform horizonal magnetic field B_0 There exists a uniform horizontal magnetic field B_0 parallel to the diameter AB of the ring Calculate tension in the OB string. The current is flowing in anticlockwise as seen from above [Given $\theta = 60^o$]



(1) $2Mg - \pi IRB_0$

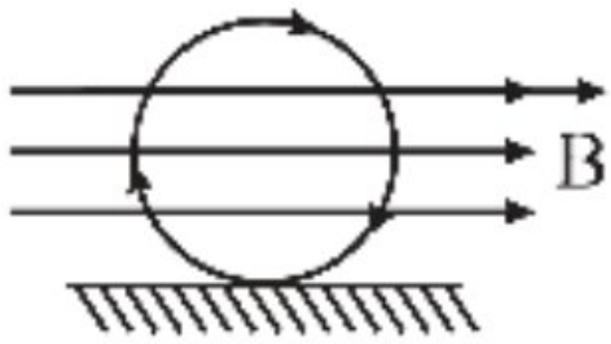
(2) $4Mg - \pi IRB_0$

(3) $6Mg - \pi IRB_0$

(4) $Mg - \pi IRB_0$

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25) A conducting ring of mass 2 kg and radius 0.5 m is placed on a smooth horizontal plane the ring carries a current i = 4A. A horizontal magnetic field B = 10 T is switched on at t = 0, the initial



angular acceleration of the ring will be

(1) 40π rad/sec^2

(2) 20π rad/sec^2

(3) 5π rad/sec^2

(4) 15π rad/sec^2

26) The work done in rotating a magnet of magnetic moment M by an angle of 90° from the external magnetic field direction is 'n' times the corresponding work done to turn it through in angle of 60°. Where 'n' gives by :-

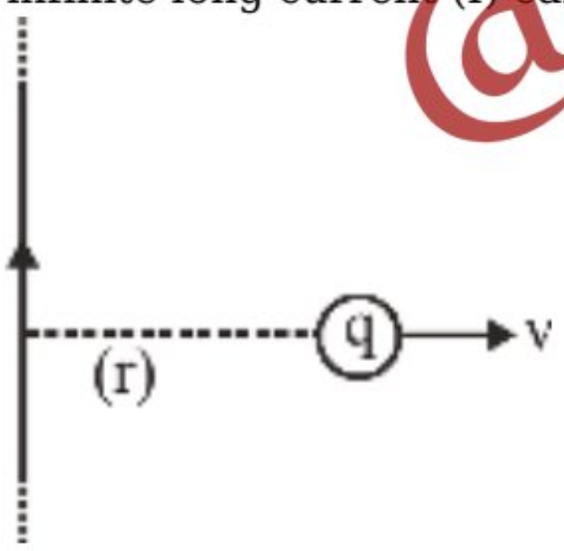
(1) $\frac{1}{2}$

(2) 2

(3) $\frac{1}{4}$

(4) 1

27) A charge (q, m) is thrown perpendicularly with speed v from a point at a distance r from an infinite long current (I) carrying wire. If its maximum distance from wire is R then :



(1) R will be infinite

(2) R will be $\left(re^{\frac{2\pi mv}{\mu_0 qI}}\right)$

(3) R will be $\left(re^{\frac{4\pi mv}{\mu_0 qI}}\right)$

(4) R will be $\left(re^{\frac{\pi mv}{\mu_0 qI}}\right)$

28)

A particle of specific charge (q/m = α) starts moving from the origin under the action of an electric

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field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{k}$. Its velocity of (x^0, y^0, 0) is $(4\hat{i} - 3\hat{j})$. Value of x_0 is

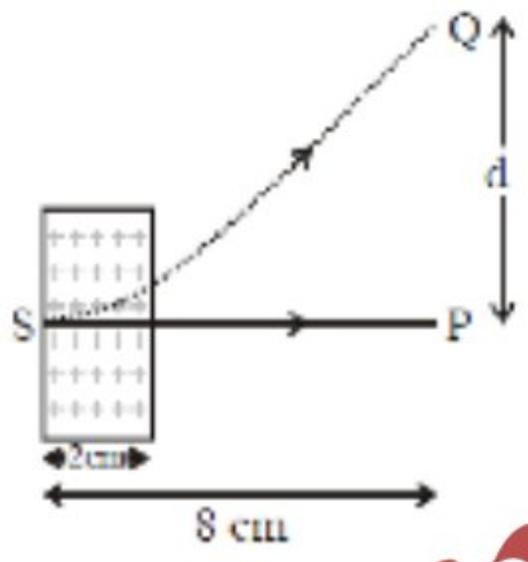
(1) $\frac{13}{2}\alpha\frac{E_0}{B_0}$

(2) $\frac{25}{2\alpha E_0}$

(3) $\frac{16\alpha B_0}{E_0}$

(4) $\frac{5\alpha}{2B_0}$

29) An electron, moving along the x-axis with an initial energy of 100 eV, enters a region of magnetic field $\vec{B} = (1.5 \times 10^{-3}T)\hat{k}$ at S (see figure). The field extends between x = 0 and x = 2 cm. The electron is detected at the point Q on a screen placed 8 cm away from the point S. The distance d between P and Q (on the screen) is : (electron's charge = 1.6×10^{-19} C, mass of electron = 9.1 × 10^{-31} kg)



(1) 1.22 cm

(2) 11.65 cm

(3) 2.25 cm

(4) 12.87 cm

30) A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E_0\hat{i}$ and $\vec{B} = B_0\hat{i}$ with a velocity $\vec{v} = v_0\hat{j}$. The speed of the particle will become $2v_0$ after a time

(1) $t = \frac{2mv_0}{qE_0}$

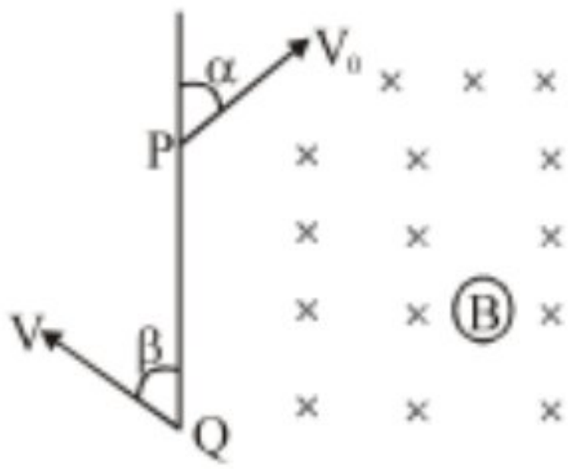
(2) $t = \frac{2B_0q}{mv_0}$

(3) $t = \frac{\sqrt{3}B_0q}{mv_0}$

(4) $t = \frac{\sqrt{3}mv_0}{qE_0}$

31) A particle of charge –q & mass m enters a uniform magnetic field B (Perpendicular to paper inwards) at P with a speed V_0 at an angle a & leaves the field at Q with speed V at angle b as shown then :-

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(1) $\alpha = \beta$

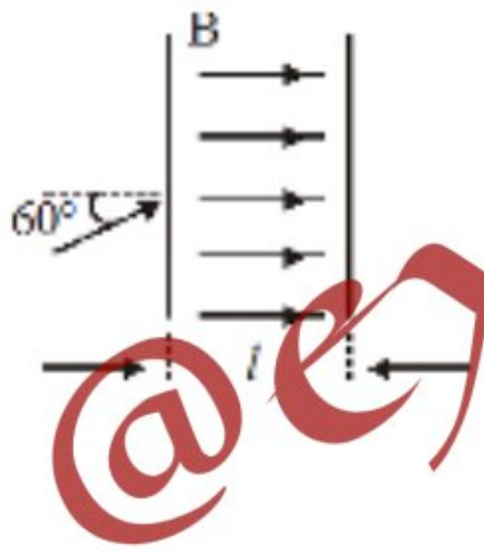
(2) $PQ = \frac{2mV_0 \sin\alpha}{qB}$

Particle remains in field for time

(3) $t = \frac{2(\pi - \alpha)m}{Bq}$

(4) All of above

32) The figure shows a region of length '□' with a uniform magnetic field of 0.3 T in it and a proton entering the region with velocity 4×10^5 ms^{-1} making an angle 60° with the field. If the proton completes 10 revolution by the time it cross the region shown, '□' is close to (mass of proton $= 1.67 \times 10^{-27}$ kg, charge of the proton $= 1.6 \times 10^{-19}$ C)



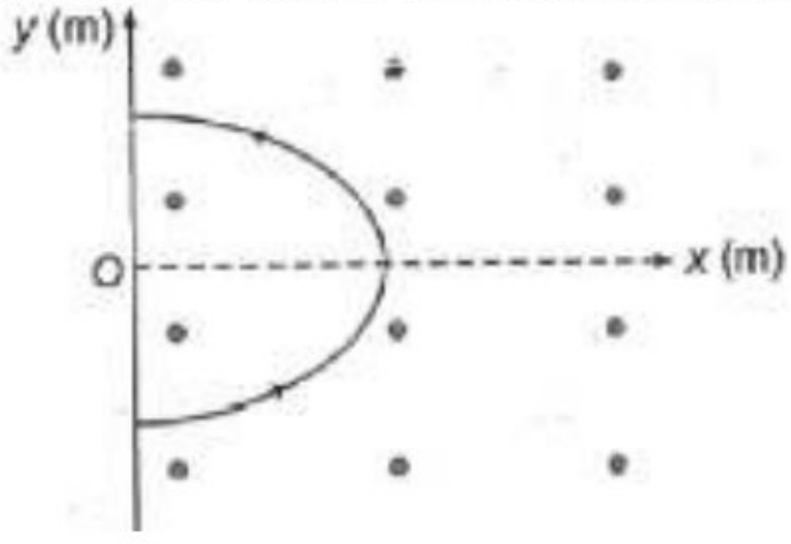
(1) 0.11 m

(2) 0.22 m

(3) 0.44 m

(4) 0.88 m

33) A wire carrying a current of 3 A is bent in the form of a parabola $y^2 = 4 - x$ as shown in figure, where x and y are in metre. The wire is placed in a uniform magnetic field $\vec{B} = 5\hat{k}$ tesla. The force acting on the wire is



(1) $60\hat{i}$ N

(2) $-60\hat{i}$ N

(3) $30\hat{i}$ N

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(4) $-30\hat{i}N$

34) What is self-inductance of coil which produces 5V when the current changes from 3 amperes to 2 amperes in one mili-second ?

(1) 5000 henry

(2) 5 milli henry

(3) 50 henry

(4) 5 henry

35) Which of the following statements is correct for diamagnetic materials :-

(1) $\mu_r < 1$

(2) □ is negative and low

(3) □ does not depend on temperature

(4) All of the above

36) A bar magnet of length 14 cm is placed in the magnetic meridian with its north pole pointing towards the geographic north pole. A neutral point is obtained at a distance of 18 cm from the center of the magnet. If $B_H = 0.4$ G, the magnetic moment of the magnet is
(1 G = 10^{-4}T)

(1) 2.880 x 10^3 J T^{-1}

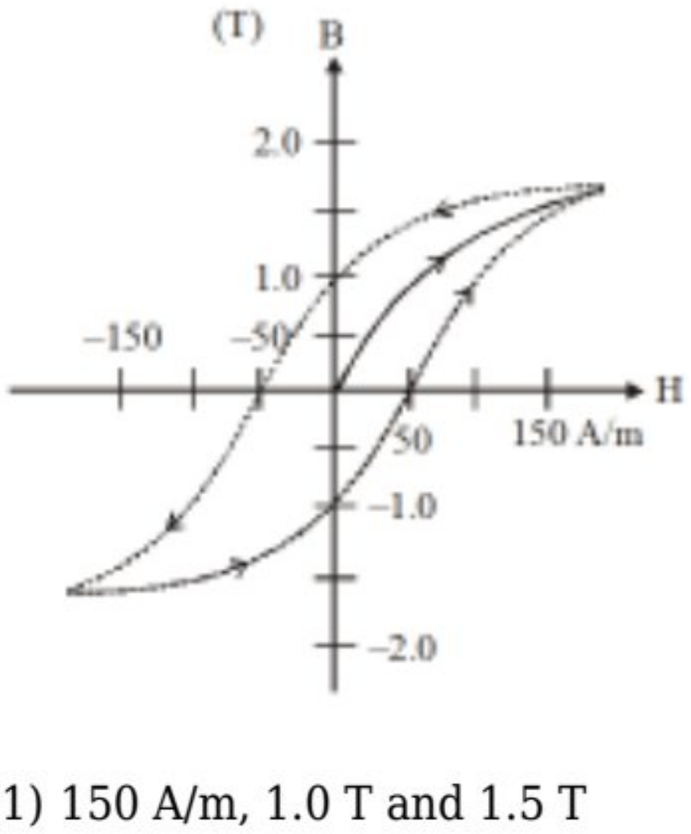
(2) 2.880 x 10^2 J T^{-1}

(3) 2.880 J T^{-1}

(4) 28.80 J T^{-1}

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37) The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are:



(1) 150 A/m, 1.0 T and 1.5 T

(2) 1.0 T, 50 A/m and 1.5 T

(3) 1.5 T, 50 A/m and 1.0 T

(4) 2.5 T, 50 A/m,1 T

38) A particle of mass m and charge q is in an electric and magnetic field is given by

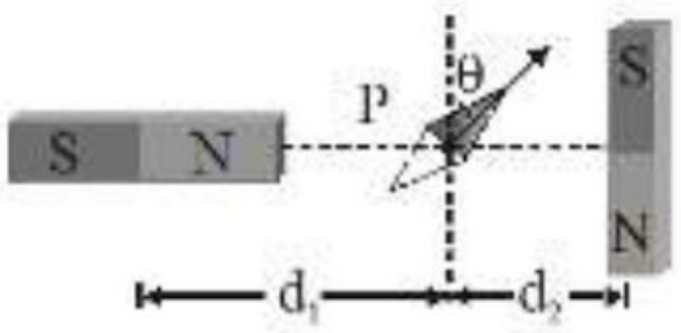
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$E = 2\hat{i} + 3\hat{j},\ B = 4\hat{j} + 6\hat{k}$.
The charged particle is shifted from the origin to the point P(x = 1; y = 1) along a straight path. The magnitude of the total work done is

(1) (0.35)q
(2) (0.15)q
(3) (2.5)q
(4) 5q

39) Two magnets A and B are identical and these are arranged as shown in the figure. Their length is negligible in comparison to the separation between them. A magnetic needle is placed between the magnets at point P which gets deflected through an angle θ under the influence of magnets. The ratio of distance d_1 and d_2 will be :-



(1) $(2\tan\theta)^{1/3}$
(2) $(2\tan\theta)^{-1/3}$
(3) $(2\cot\theta)^{1/3}$
(4) $(2\cot\theta)^{-1/3}$

40) A certain amount of current when flowing in a properly set tangent galvanometer, produces a deflection of 45°. If the current be reduced by a factor of $\sqrt{3}$, the deflection would -

(1) Decrease by 30°
(2) Decrease by 15°
(3) Increase by 15°
(4) Increase by 30°

41) Magnetic field produced by eletrons in atoms and molecules is due to their

(1) spin motion only
(2) orbital motion only
(3) spin and orbital motion both
(4) neither spin nor orbital motion

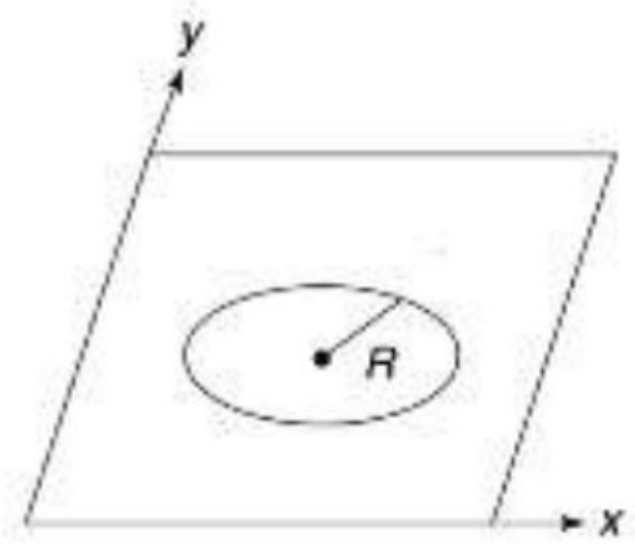
42) A vibration magnetometer consists of two identical bar magnets placed one over the other such that they are perpendicular and bisect each other. The time period of oscillation in a horizontal magnetic field is $(2)^{5/4}$ seconds. One of the magnets is removed and if the other magnet oscillates in the same field, then the time period in seconds is:-

(1) $2^{1/4}$
(2) $2^{1/2}$
(3) 2

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(4) $2^{3/4}$

43) A conducting ring of mass $m = \pi\,kg$ and radius $R = \frac{1}{2}m$ is kept on a flat horizontal surface (xy plane). A uniform magnetic field is switched on in the region which changes with time (t) as $\vec{B} = \left(2\hat{j} + t^2\hat{k}\right)$ T. Resistance of the ring is $r = \pi\Omega$, $g = 10\,ms^{-2}$
Calculate the induced electric field at the circumference of the ring at the instant it begins to topple.



(1) 10 V/m

(2) 20 V/m

(3) 36 V/m

(4) 26 V/m

44) Given below are two statements:
Statement I: The electric force changes the speed of the charged particle and hence changes its kinetic energy: whereas the magnetic force does not change the kinetic energy of the charged particle.
Statement II: The electric force accelerates the positively charged particle perpendicular to the direction of electric field. The magnetic force accelerates the moving charged particles along the direction of magnetic field.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

45) A proton with a kinetic energy of 2.0 eV moves into a region of uniform magnetic field of magnitude $\frac{\pi}{2} \times 10^{-3}$ T. The angle between the direction of magnetic field and velocity of proton is 60°. The pitch of the helical path taken by the proton is _____cm.
(Take, mass of proton = 1.6×10^{-27} kg and charge on proton = 1.6×10^{-19} C).

(1) 20

(2) 40

(3) 60

(4) 80

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CHEMISTRY

1) Which of the following is related to Nessler's reagent?

(1) $PtCl_4 + KCl \rightarrow$

(2) $AgCl + NH_3 \rightarrow$

(3) $AgBr + Na_2S_2O_3 \rightarrow$

(4) $HgI_2 + KI \rightarrow$

2) When a mixture of solid NaCl, solid $K_2Cr_2O_7$ is heated with concentrated H_2SO_4, orange red vapours are obtained.These are of the compound :-

(1) chromous chloride

(2) chromyl chloride

(3) chromic chloride

(4) chromic sulphate

3) An aqueous solution of gas (X) turns red litmus paper into blue; then gas will be :

(1) SO_2

(2) NO_2

(3) NH_3

(4) CO_2

4) The compound present in borax bead is

(1) B_2O_3

(2) $NaBO_2$

(3) $NaBO_3$

(4) $NaBO_2 + B_2O_3$

5) Dil. $HCl + H_2S$ is the group reagent of :

(1) I^{st} group

(2) IInd group

(3) IIIrd group

(4) IVth group

6) H_2S on passing through ammoniacal solution gives white ppt. The white ppt. is of :

(1) CoS

(2) NiS

(3) MnS

(4) ZnS

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7) Which does not dissolve in concentrated ammonium hydroxide solution ?

(1) AgF
(2) AgCl
(3) AgBr
(4) AgI

8) Which cation is detected by flame test?

(1) NH_4^+
(2) K^+
(3) Mg^{+2}
(4) Al^{+3}

9) Few drops of a salt solution are shaken with chloroform, chlorine water. Chloroform layer becomes violet. Solution contains :

(1) NO_2^- ion
(2) NO_3^- ion
(3) Br^- ion
(4) I^- ion

10) **Statement-I :** The chlorides of Be^{2+} and Mg^{2+} do not impart any colour to the flame test.
Statement-II : The hydration energy of Be^{2+} is higher than that of Mg^{2+}.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct
(4) Both **Statement I** and **Statement II** are correct.

11) What is the colour of $CoO.Al_2O_3$

(1) Pink
(2) Blue
(3) Bluish white
(4) Green

12) Nessler reagent is used to detect.

(1) NH_4^+
(2) PO_4^{-3}
(3) MnO_4^-
(4) SO_4^{-2}

13) The salt which undergoes anionic hydrolysis only is

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(1) $CuSO_4Na$

(2) NaCl

(3) CH_3COONa

(4) $HCOONH_4$

14) The metal that does not give the borax bead test is :-

(1) Cr

(2) Ni

(3) Na

(4) Mn

15) Ferric ion forms a following prussian blue coloured ppt of :-

(1) $K_4[Fe(CN)_6]$

(2) $Fe_4[Fe(CN)_6]_3$

(3) $KMnO_4$

(4) $Fe(OH)_3$

16) MnO_2 and H_2SO_4 added to NaCl, the greenish yellow gas liberated is

(1) Cl_2

(2) NH_3

(3) N_2

(4) H_2

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17) For a catalyst which condition is not essential :

(1) Variable oxidation states

(2) High ionisation energy

(3) Vacant orbitals

(4) Free valency on the surface

18) $KMnO_4$ on treatment with conc. H_2SO_4 forms a compound (X) which decomposes explosively on heating forming (Y). The (X) and (Y) are respectively

(1) Mn_2O_7, MnO_2

(2) Mn_2O, Mn_2O_3

(3) $MnSO_4$, Mn_2O_3

(4) Mn_2O_3, MnO_2

19) $(NH_4)_2Cr_2O_7$ (Ammonium dichromate) is heated. The green coloured powder blown in air is :

(1) Cr_2O_3

(2) CrO_2

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(3) Cr_2O_4

(4) CrO_3

20) The higher oxidation states of transition elements are found to be the combination with A and B, which are:

(1) F, O

(2) O, N

(3) O, Cl

(4) F, Cl

21) Volatile metals are :-

(1) Cu, Ag, Au

(2) Zn, Cd, Hg

(3) C, Pb

(4) Fe, Co, Ni

22) When MnO_2 is fused with KOH in the presence of O_2 a coloured compound is formed. Compound & the colour of compound is :-

(1) K_2MnO_4, green

(2) $KMnO_4$, purple

(3) Mn_2O_3, brown

(4) Mn_2O_4, black

23)

Four successive members of the first row transition elements are listed below with atomic numbers. Which one of them is expected to have the highest oxidising character ($E^0_{M^{3+}/M^{2+}}$) ?

(1) Cr(Z = 24)

(2) Mn(Z = 25)

(3) Fe (Z = 26)

(4) Co(Z = 27)

24) $Cu^{+2}_{(aq)}$ is more stable than $Cu^{+}_{(aq)}$, because :-

(1) IP_2 is less than IP_1

(2) $3d^9$ is more symmetric than $3d^{10}$

(3) $\Delta H_{hydration}$ of Cu^{+2} is greater than $\Delta H_{hydration}$ of Cu^{+}

(4) $\Delta H_{hydration}$ of Cu^{+} is greater than $\Delta H_{hydration}$ of Cu^{+2}

25) Which of the following increasing order of oxidising power is correct for the following species ?

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VO_2^+, MnO_4^-, $Cr_2O_7^{2-}$

(1) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$

(2) $VO_2^+ < MnO_4^- < Cr_2O_7^{2-}$

(3) $Cr_2O_7^{2-} < VO_2^+ < MnO_4^-$

(4) $Cr_2O_7^{2-} < MnO_4^- < VO_2^+$

26) When hydrogen peroxide is added to acidified potassium dichromate, a blue colour is produced due to formation of :-

(1) CrO_3

(2) $Cr_2O_7^{-2}$

(3) CrO_5

(4) CrO_4^{-2}

27) Select the wrong statement:

(1) Eu^{2+} and Yb^{2+} ions are more stable due to half-filled and fully filled f-orbitals

(2) Ce^{4+} is a stable oxidation state in lanthanides

(3) All lanthanides predominantly show +2 oxidation state

(4) Gd^{3+} shows extra stability due to half-filled f^7 configuration

28) Blue vitrol contains which type of water molecules :-

(1) H-bonded, interstitial

(2) H-bonded, co-ordinated

(3) Interstitial only

(4) Co-ordinated only

29) CrO_4^{2-} (Yellow) changes to $Cr_2O_7^{2-}$ (orange) in pH = x and vice versa in pH = y, Hence x and y are :-

(1) 6, 8

(2) 6, 5

(3) 8, 6

(4) 7, 7

30) Which of the following species can act as an oxidising agent ?

(1) Yb^{+2}

(2) Eu^{2+}

(3) Ce^{+4}

(4) La^{+3}

31) Select the incorrect statement about $K_2Cr_2O_7$:-

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(1) the bond angle of Cr–$\hat{O}$–Cr > 109.5

(2) the hybridisation is d^3s

(3) two types of Cr–O bond length are present

(4) It is acts as redusing agent.

32) Electronic configuration of Th is :-

(1) [Rn] $5f^1,6d^1,7s^2$

(2) [Rn] $5f^0,6d^2,7s^2$

(3) [Rn] $5f^2,6d^0,7s^2$

(4) [Xe] $5f^0,6d^2,7s^2$

33) I^- can't exist with :-

(a) Cu^+ (b) Pb^{+2} (c) Tl^{+3} (d) Bi^{+5}

(e) Cu^{+2} (f) Pb^{+4} (g) Tl^{+1}

Correct option is :

(1) b, c, d, f

(2) c, d, e, f

(3) b, d, e, f

(4) a, c, d, e

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34) The number of faradays required to reduce 5 gram equivalent of Ni^{+2} to Ni metal will be :

(1) 1

(2) 2

(3) 0.5

(4) 5

35) Which one will give Br_2 gas by reaction with KBr solution :-

(1) N_2

(2) I_2

(3) Cl_2

(4) NO_2

36) What products will be formed at anode and cathode respectively by electrolysis of aqueous NaCl in presence of Hg cathode ?

(1) Cl_2, H_2

(2) O_2, H_2

(3) O_2, Cl_2

(4) Cl_2, Na

37) Which has maximum conductivity ?

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(1) 0.1 m HCl

(2) 0.1 m NaCl

(3) 0.1 m CH_3COOH

(4) H_2O

38) Which of the following is not a function of salt bridge in an electrochemical cell ?

(1) To maintain electrical neutrality of the solution

(2) To complete the circuit so that current can flow.

(3) To minimize liquid-liquid junction potential

(4) To increase cell voltage

39) The equation representing the process by which standard reduction potential of zinc can be defined is

(1) $Zn^{2+}(s) + 2e^- \rightarrow Zn(s)$

(2) $Zn(g) \rightarrow Zn^{2+}(g) + 2e^-$

(3) $Zn^{2+}(g) + 2e^- \rightarrow Zn(s)$

(4) $Zn^{2+}(aq.) + 2e^- \rightarrow Zn(s)$

40) The highest electrical conductivity of the following aqueous solutions is of :-

(1) 0.1 M acetic acid

(2) 0.1 M chloroacetic acid

(3) 0.1 M fluroacetic acid

(4) 0.1 M difluoroacetic acid

41) The metal that can not be obtained on reduction of its oxide by aluminium is –

(1) K

(2) Mn

(3) Cr

(4) Fe

42) When Ionic hydrides are electrolysed H2 gas is obtained at ?

(1) Cathode

(2) Anode

(3) Both

(4) None

43) Which has maximum conductivity:

(1) $[Cr(NH_3)_3Cl_3]$

(2) $[Cr(NH_3)_4Cl_2]Cl$

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NOVA

(3) $[Cr(NH_3)_5Cl]Cl_2$

(4) $[Cr(NH_3)_6]Cl_3$

44) For electrolysis match column-I with column-II,

	Column-I		**Column-II**
(A)	Very dilute solution of NaCl	(p)	O_2 evolved at anode
(B)	Very dilute solution of HCl	(q)	H_2 evolved at cathode
(C)	Concentrated solution of NaCl	(r)	Cl_2 evolved at anode
(D)	Fairly concentrated solution of $AgNO_3$	(s)	Ag deposited at cathode

(1) A → (p,q), B → (p,q), C → (q,r), D → (p,s)

(2) A → (q,r), B → (q,r), C → (p,q), D → (p,s)

(3) A → (p,q), B → (q,r), C → (p,q), D → (p,s)

(4) A → (q,r), B → (p,q), C → (p,q), D → (p,s)

45) The efficiency of a fuel cell is given by

(1) $\frac{\Delta H}{\Delta G}$

(2) $\frac{\Delta G}{\Delta S}$

(3) $\left|\frac{\Delta G}{\Delta H}\right|$

(4) $\frac{\Delta S}{\Delta G}$

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BIOLOGY

1) Match the columns with regard to innate immunity :-

Column I		**Column II**	
(A)	Interferons	(i)	Physiological barrier
(B)	Gastric juice, saliva, tear and sweat	(ii)	Physical barrier
(C)	Mucosa and skin	(iii)	Cellular barrier
(D)	Neutrophils, monocyte, N-K cell	(iv)	Cytokine barrier

(1) A - i, B - ii, C - iii, D - iv

(2) A - iv, B - i, C - ii, D - iii

(3) A - iv, B - i, C - iii, D - ii

(4) A - iii, B - iv, C - i, D - ii

2) Match the column-I with column-II and choose the correct options :-

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Column-I (Disease)		Column-II (Antibodies formed against)	
(A)	Hashimoto's Disease	(i)	Castle intrinsic factor
(B)	Myasthenia gravis	(ii)	Thyroid gland
(C)	Pernicious Anemia	(iii)	β-cells of pancreas
(D)	Insulin dependent diabetes mellitus (I.D.D.M.)	(iv)	Acetylcholine receptors

(1) A-iii, B-ii, C-iv, D-i

(2) A-iv, B-iii, C-i, D-ii

(3) A-ii, B-iv, C-i, D-iii

(4) A-ii, B-i, C-iv, D-iii

3) Which of the following correctly defines antigens and antibodies?

(1) Antigens are foreign molecules recognized by the immune system and trigger antibody production; antibodies are specialized glycoproteins produced by plasma cells that bind specifically to antigens and mediate immune defense.

(2) Antigens are molecules produced by the body to fight infections; antibodies are foreign substances that trigger immune responses.

(3) Antibodies are types of antigens found in vaccines; antigens are protective proteins produced by the immune system.

(4) Antigens and antibodies are both toxic molecules that damage pathogens directly without immune involvement.

4) Each immunoglobin has two heavy chains & two light chains, the antigen binding site is present in

(1) Variable region of heavy chain

(2) Variable region of both heavy and light chain

(3) Variable region of light chain

(4) Constant region of both light and heavy chain



5) Which of the following is correct about this plant and its product? (a) Datura

(b) Hallucinogen

(c) Heroin

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(d) White, odourless, bitter, crystalline compound
(e) Interfere with the transport of dopamine
(f) Diacetyl morphine
(g) *Papaver somniferum*

(1) a, b, e

(2) d, e, f, g

(3) c, d, f, g

(4) a, c, d, e

6) Fill in the blanks and choose the correct option
(A) ___(A)___was a typhoid carrier who continued to spread typhoid for several years.
(B) ___(B)___ is responsible for the disease pneumonia in humans.
(C) ___(C)___ is one of the most common infectious diseases in man.
(D) Pathogens that enter the ___(D)___ must know a way of surviving in the stomach at low pH.

	(A)	(B)	(C)	(D)
(1)	Mary Mallon	*Salmonella typhi*	Common cold	Gut
(2)	Mary Mallon	*Haemophilus influenzae*	Common cold	Respiratory tract
(3)	Mary Mallon	*Haemophilus influenzae*	Ringworms	Gut
(4)	Mary Mallon	Influenzae virus	Ringworms	Gut

(1) 1

(2) 2

(3) 3

(4) 4

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7) Find out the wrongly matched pair from the given below option.

(A)	Graft rejection	Humoral immune response
(B)	Interferon	Antiviral protein secreted by virus infected cell & cytokine barrier
(C)	Hepatitis B vaccine	Recombinant RNA technology
(D)	Macrophage and PMNL-Neutrophils	Phagocytic cells

(1) A and C

(2) A, B and C

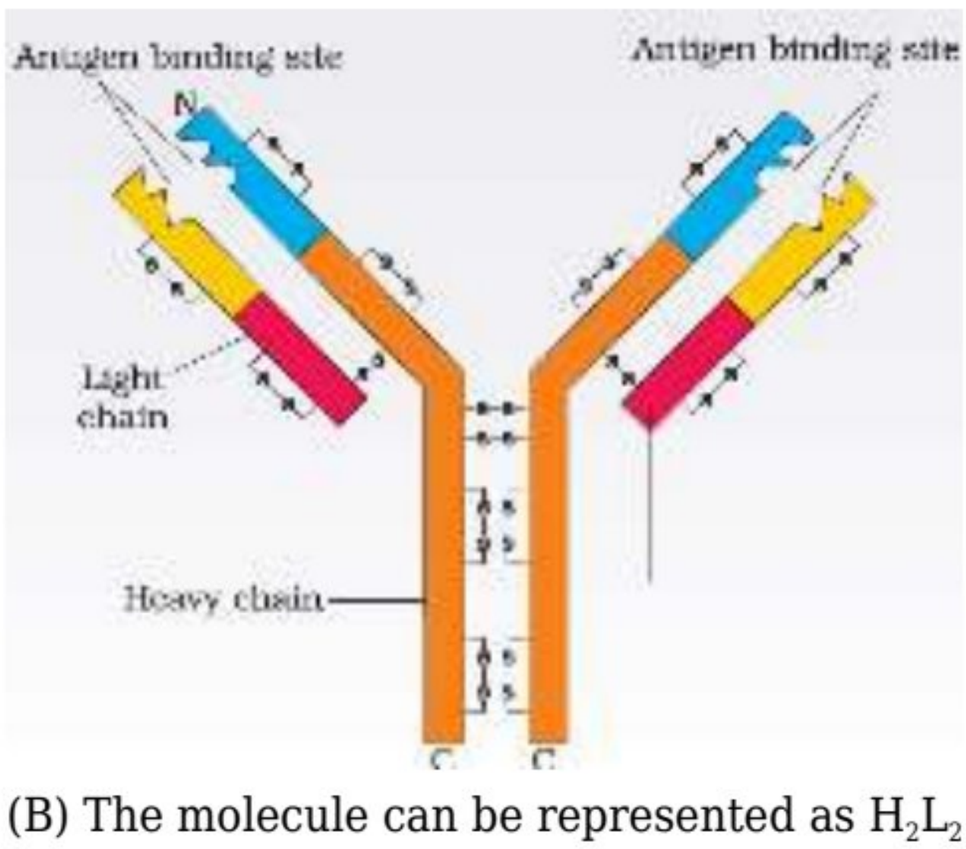
(3) Only A

(4) B and C

8) Choose the correct statements regarding the structure given in the below diagram.

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(A) The given structure is of a glycoprotein molecule.
(B) The molecule can be represented as H_2L_2 molecule.
(C) The given molecule comprises major part of our innate immunity.
(D) The molecule is synthesized by T-lymphocytes in our body.
(E) The response generated by this molecule is called as humoral immune response.

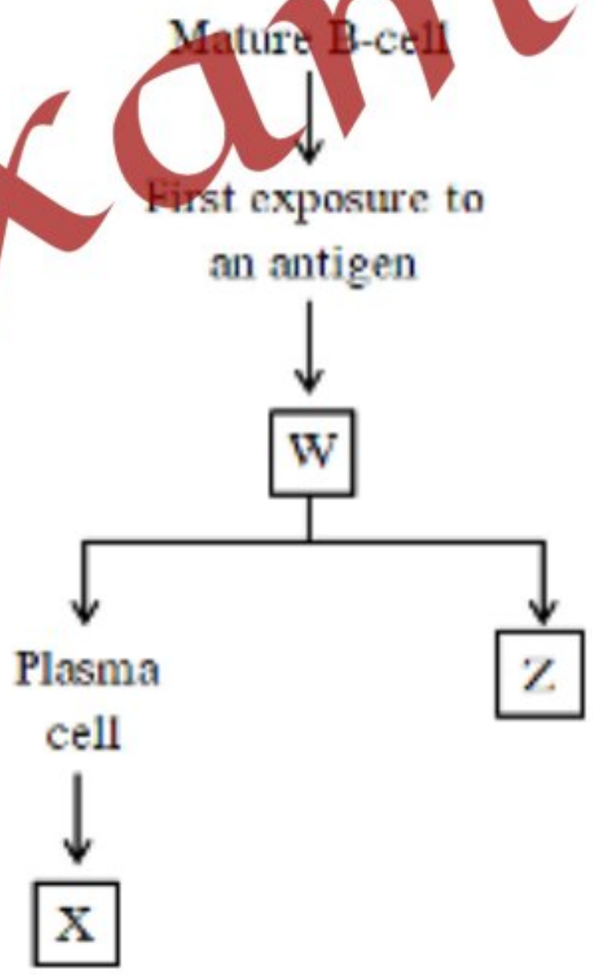
(1) A, B, C, D and E

(2) A, B, C and E

(3) A, B, D and E

(4) A, B and E

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9) Identify W, X and Z in flow chart?

W = clones of B cells.
(1) X = Antibody
Z = memory cell

W = clones of B cells.
(2) Z = Antibody
X = Memory cell

W = Antibody,
(3) Z = Clones of B-cells.
X = Memory cell

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W = Memory cell
(4) Z = Antibody
X = Antigen

10) Which of the following is example of active immunity :

(1) Foetus receives antibodies from mother through placenta during pregnancy

(2) Colostrum secreted by mother during initial days of lactation has abundant antibodies to protect the infant

(3) Injecting the microbes deliberately during immunisation

(4) All of the above

11) Choose the incorrectly matched option from the following table :

	Disease	**Pathogen**	**Symptom**
(1)	Typhoid	Salmonella	Stomach pain
(2)	malaria	Plasmodium	Chills and high fever due to hemozoin
(3)	Ascariasis	Common round worm	Blockage in intestine passage
(4)	Ringworm	Worms	Dry, scaly, itchy lesions on skin

(1) 1
(2) 2
(3) 3
(4) 4

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12) Fill in the blanks
(i) In __(a)__, a piece of the suspected tissue cut into thin section is stained and examined under microscope __(b)__ by a pathologist.
(ii) In MRI __(c)__ are used.
(iii) In CT scan __(d)__ are used.

(1) a - biopsy
b - histopathological studies
c - x-rays
d - strong magnetic field

(2) a - histopathological studies
b - biopsy
c - gamma rays
d - x-rays

(3) a - biopsy
b - histopathological studies
c - strong magnetic field
d - x-rays

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(4)
a - Histopathological studies
b - biopsy
c - x-rays
d - gamma rays

13) (i) Thymus (ii) Spleen
(iii) Appendix (iv) MALT
(v) Bone marrow
Which of the above are not secondary lymphoid organs ?

(1) iii, iv only
(2) i and v
(3) i, ii, iii. iv. v
(4) i, ii, iii, iv only

14)

Which of the following are Autoimmune disorders?
A. Myasthenia gravis
B. Rheumatoid arthritis
C. Gout
D. Muscular dystrophy
E. Systemic Lupus Erythematosus (SLE)
Choose the most appropriate answer from the options given below:

(1) A, B & D only
(2) A, B & E only
(3) B, C & E only
(4) C, D & E only

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15) Read the following statements :-
(A) AIDS is congenital disease.
(B) Sloughed cell of malignant tumour reaches distant site this is called metastasis.
(C) Biopsy uses strong magnetic field to detect pathological changes in living tissue.
(D) Proto oncogene is responsible for abnormal cellular division.
How many of above statements are correct ?

(1) 4
(2) 2
(3) 3
(4) 1

16) How many statements are incorrect from given below ?
(a) Low intensified immune response is anamnestic response.
(b) High intensified response is called as secondary immune response .
(c) A.T.S. contains ready made antibodies against tetanus.
(d) Allergic antibodies are IgE type.

(1) One

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(2) Two

(3) Three

(4) Four

17) Match the column (A) with (B)

	Column-A		Column-B
(i)	Allergy	(a)	IgM
(ii)	Exocrine secretions	(b)	IgG
(iii)	Universal	(c)	IgA
(iv)	Pentameric	(d)	IgD
(v)	B-cell surface receptor	(e)	IgE

	(i)	(ii)	(iii)	(iv)	(v)
(1)	a	b	c	d	e
(2)	e	c	b	a	d
(3)	d	a	b	c	e
(4)	c	e	a	d	b

(1) 1

(2) 2

(3) 3

(4) 4

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18)

Withdrawal syndrome is characterized by :-
(i) Anxiety (ii) Violence
(iii) Shakiness (iv) Sweating

(v) Anemia (vi) Nausea

(1) i, ii, iii and iv

(2) i, ii and iii

(3) i, iii, iv and v

(4) i, iii, iv and vi

19) Identify the correct statements from the following?
(a) Typhoid is the vector born disease.
(b) T-helper cells are called as HIV factory
(c) Primary immune response is of low intensity and slow
(d) Hepatitis B and AIDS are example of sexually transmitted disease
(e) Down syndrome is the most common sex trisomy
(f) Characters of albinism also found in Alkaptonuria & Phenyl ketonuria

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(1) c, d, e, f

(2) a, b

(3) b, c, d, e

(4) c, d

20)

Match these columns :-

Column-I		Column-II	
A	*Rhino virus*	i	Female culex
B	*Heamophilus influenzae*	ii	Dysentery
C	*Wuchereria bancrofti*	iii	Infect the alveoli of the lungs
D	*Entamoeba histolytica*	iv	Usually last for 3-7 days

(1) A-iv, B-iii, C-ii, D-i

(2) A-iv, B-iii, C-i, D-ii

(3) A-iii, B-iv, C-i, D-ii

(4) A-iii, B-ii, C-iv, D-i

21) Cancer will appear when

(a) cells looses property of contact inhibition

(b) oncogenic virus enters in body

(c) proto oncogene activates in cells

(d) body is exposed to ionising radiations for long time

(e) benign tumors appear in body

(1) a, b, c

(2) a, b, d

(3) a, b, c, d

(4) only c, e

22) Burkitt's lymphoma is caused due to reciprocal translocation between chromosome :-

(1) 22 and 9

(2) 22 and 14

(3) 8 and 14

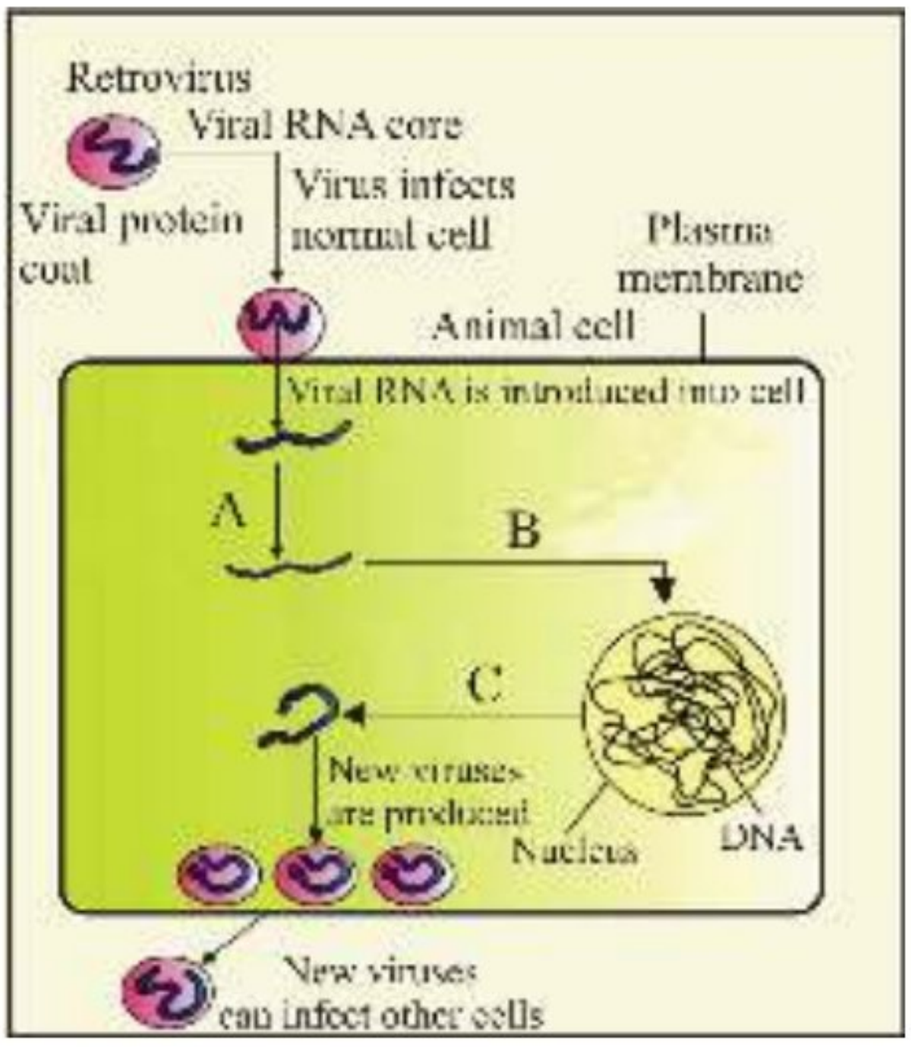
(4) 9 and 14

23) The figure below shows steps A, B & C of replication of retrovirus. Select the option giving

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correct identification

(1) B–Incorporation of viral DNA into host genome with integrase enzyme

(2) C–Production of new viral DNA by infected cell

(3) A–Replication of viral RNA by protease enzyme

(4) C–Viral RNA replication by reverse transcriptase enzyme

24) True about nicotine is :-
(a) Adrenal gland stimulant
(b) Alkaloid
(c) Found in tobacco
(d) Responsible for addiction
(e) Powerful vasodialator

(1) a, b, d, e

(2) b, c, d, e

(3) a, b, c, d

(4) a, b, c, d, e

25)

Auto antibodies formed against our self antigen is auto immune disorders. Match the following column-A & B by keep in this in consideration?

	A		B
(i)	Myasthenia Gravis	(a)	Myelin sheath
(ii)	Pernicious Anemia	(b)	β-cells of pancreas
(iii)	IDDM	(c)	Intrinsic castle factor

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(iv)	Multiple sclerosis	(d)	Acetyl choline receptor

	(i)	(ii)	(iii)	(iv)
(1)	d	c	b	a
(2)	d	c	a	b
(3)	c	d	b	a
(4)	c	b	a	d

(1) 1
(2) 2
(3) 3
(4) 4

26) Recognise the figure and find out the correct matching:-


a


b

(1) 'a' produces depressant while 'b' influences cardiovascular system
(2) Receptors of 'a' are present in CNS and GIT while for 'b' are present principally in brain.
(3) 'a' produces smack while 'b' produces marijuana
(4) All of the above

27) Which of the following is not related to alcoholism:-
(i) Amnesia (ii) Gastritis
(iii) Lung Cancer (iv) Liver Cirrhosis
(v) Fatty liver syndrome (vi) Bronchitis

(1) iii, iv, v, vi
(2) i, ii, iv, v, vi
(3) iii, vi
(4) i, ii, iii, iv, v, vi

28) A child living in a coastal village shows the following conditions :-
• Swollen limbs
• Gross deformities in genital organ
• Blockage in lymphatic vessels observed in medical examination.
This condition is most likely caused due to infection by :-

(1) Female anopheles mosquito carrying plasmodium.
(2) Culex mosquito carrying wuchereria.

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(3) Sandfly carrying leishmania donovani.

(4) House fly carrying parasitic worm.

29) Read the following statement (A-D) :
(A) There is always a time-lag between the infection and appearance of AIDS symptoms
(B) AIDS is caused by HIV, a member of a group of viruses called retrovirus
(C) AIDS does not spread by mere touch or physical contact, it spreads only through body fluids
(D) Transmission of HIV infection can only take place through sexual contact and by no other means
How many of the above statements are **correct**?

(1) 4

(2) 1

(3) 2

(4) 3

30) In the given list how many disease are Auto immune?
SCID, Bronchial asthma, Pernicious anemia, Rheumatoid arthritis, Multiple sclerosis

(1) 3

(2) 4

(3) 5

(4) 2

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31) J-chain is a polypeptide chain associated with which of the following immunoglobulins?

(1) IgG only

(2) IgG & IgM

(3) IgA only

(4) IgM & IgA

32) The main reason why antibiotics could not solve all the problems of bacteria mediated diseases is

(1) insensitivity of the individual following prolonged exposure to antibiotics

(2) inactivation of antibiotics by bacterial enzymes

(3) decreased efficiency of the immune system

(4) the development of mutant strains resistant to antibiotics

33) How many of the following are bacterial diseases :-
(a) Diphtheria (b) Dengue (c) Typhoid
(d) Filariasis (e) Tuberculosis (f) Tetanus

(1) Four

(2) Three

(3) Two

(4) One

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34) Common cold differs from pneumonia in, that :

(1) Pneumonia is caused by a virus while the common cold is caused by the bacterium Haemophilus influenzae

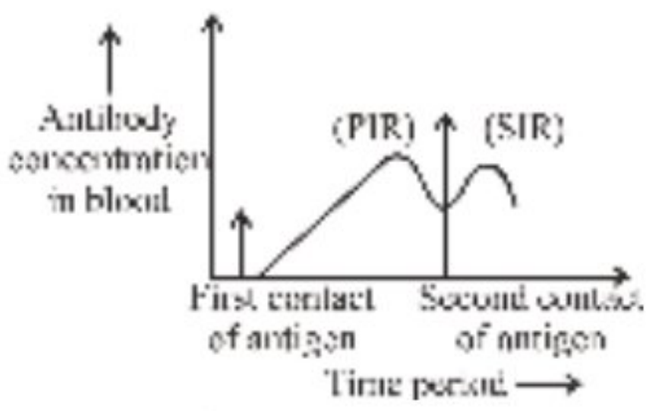
(2) Pneumonia pathogen infects alveoli whereas the common cold affects nose and respiratory passage but not the lungs.

(3) Pneumonia is a communicable disease whereas the common cold is a nutritional deficiency disease.

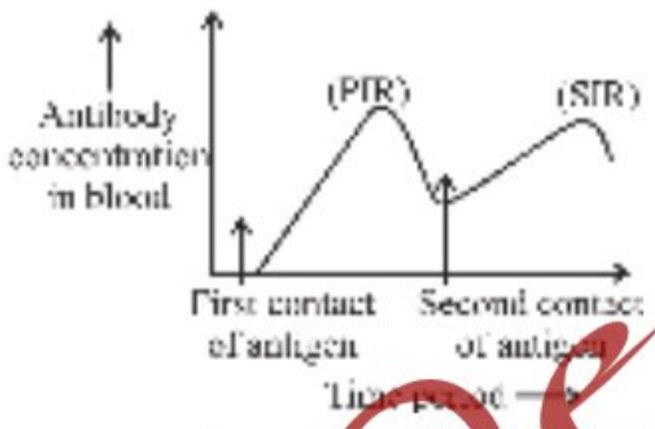
(4) Pneumonia is always prevented by a live attenuated bacterial vaccine whereas the common cold has no effective vaccine.

35) Which of the following graphs is correct representation for primary and secondary immune responses?

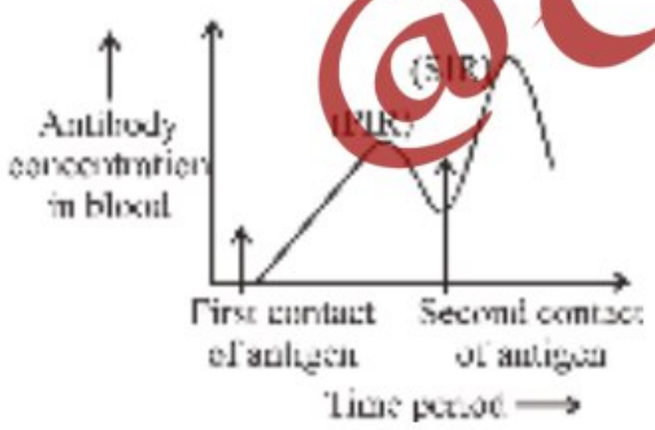
(1)



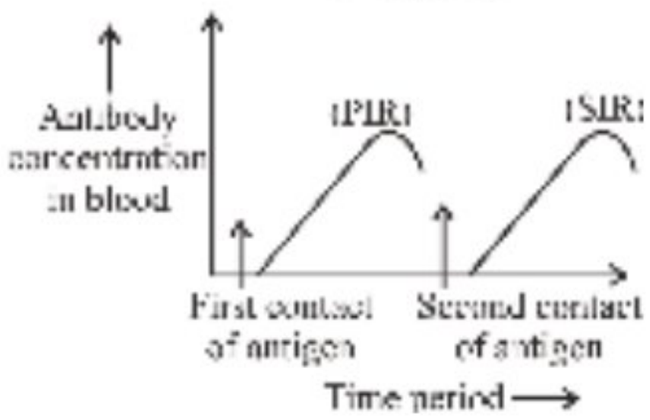
(2)



(3)



(4)



36) How many secondary lymphoidal organs are here?
Bone-marrow, Spleen, Thymus, Lymph node, Thyroid, Peyer's patches, Kupffer cells

(1) 2

(2) 3

(3) 4

(4) 5

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37) How many of the following are required for organ transplantation in case of autograft.
Blood group matching, Immuno potentiation drugs, MHC-matching, Vaccination

(1) Zero

(2) One

(3) Two

(4) Three

38) Person having intermittent rhinitis problem. After checkup, doctor prescribed differential antibody concentration count. Concentration of IgE remarkably shoot up from normal in that person. Most probable diagnosis may be ?

(1) Skin cancer

(2) Goitre

(3) Hay fever

(4) Enteric fever

39) Drugs used to reduce the symptoms of allergy :-
(i) Antihistaminic drugs
(ii) Adrenaline
(iii) Steroids
(iv) α-interferon

(1) i, ii and iv

(2) i, ii and iii

(3) ii and iii

(4) i and ii

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40)

At new place you suddenly starts sneezing and wheezing.
The statement correctly correlate this condition
(i) This is exaggerated response of the immune system
(ii) Symptoms are watery eyes and running nose
(iii) Due to the release & chemicals like adrenaline, steroids and histamine

(iv) Antibodies produced to these are IgE type (v) Rheumatoid arthritis.

(1) i, ii, iii iv are false

(2) i, iii, iv are true

(3) i, ii, iv are true

(4) all are correct except (v)

41)

How many of the following is not a congenital disease?

Down’s syndrome, Cry du chat syndrome, Pneumonia, Typhoid, AIDS, Amoebiasis.

(1) 4

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(2) 3

(3) 2

(4) 5

42)

Match the following :

Column I		Column II	
1	Autograft	A	Transplantation between same genetic Individual
2	Isograft	B	Transplantation between different genetic person
3	Allograft	C	Transplantation of tissue from one body part to another of an Individual
4	Xenograft	D	Transplantation between different species

(1) 1-C, 2-B, 3-D, 4-B

(2) 1-C, 2-A, 3-D, 4-B

(3) 1-C, 2-A, 3-B, 4-D

(4) 1-B, 2-D, 3-A, 4-C

43) How many disease in the list given below are transmitted through droplet infection :
Tuberculosis, Pertusis, Dengue, Filariasis, Cholera, Polio, Influenza, Rabies, Hepatitis, Common cold, Pneumonia.

(1) 6

(2) 4

(3) 5

(4) 1

44) Dysentery, Amoebiasis, Ascariasis, Filariasis, Hepatitis B, Typhoid, Diphtheria, Pneumonia
How many of the following spreads through contanimated food and water :-

(1) Three

(2) Five

(3) Four

(4) Six

45) How many of the given statements are correct?
a. Intravenous drug abusers are at a high risk of acquiring HIV infection.
b. Chemical carcinogens like UV rays and X-rays cause neoplastic transformation.
c. Cancer is the most dreaded infectious disease in humans.
d. Drugs used in chemotherapy besides targeting cancer cells, also inhibit the proliferation of normal cells in the body leading to hair loss and anaemia.

(1) Four

(2) Three

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(3) Two

(4) One

46) A hexaploid female plant (6n) is artificially crossed with a diploid male plant (2n). Assuming successful fertilization, what will be the ploidy of the Primary Endosperm Nucleus (PEN) and the zygote, respectively?

(1) 4n and 5n

(2) 7n and 4n

(3) 8n and 4n

(4) 5n and 4n

47) If a chemical specifically inhibits the synthesis of the 'callase' enzyme in the anthers of an angiosperm, which of the following events will be directly and immediately halted?

(1) Meiosis in the Microspore Mother Cell

(2) Separation of microspore tetrads into individual pollen grains

(3) Formation of sporopollenin in the exine

(4) Mitotic division of the generative cell into two male gametes

48) An angiosperm plant with the genotype AaBbCc undergoes apomictic seed development exclusively via adventive polyembryony. What will be the genotype of the resulting embryos?

(1) AABBCC or aabbcc

(2) ABC or abc

(3) AaBbCc

(4) Highly variable due to meiotic recombination

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49) During the entry of the pollen tube into the embryo sac, which of the following sequences most accurately describes the path and immediate post-entry events?

(1) Passes directly through the egg cell → discharges two male gametes into the central cell

(2) Enters one synergid → synergid degenerates → male gametes released into the synergid cytoplasm

(3) Bypasses the synergids completely → enters central cell → releases gametes

(4) Enters via antipodal cells → travels to the micropylar end → fertilizes the egg

50) To produce 400 viable seeds in a typical dicot plant, what is the absolute minimum number of meiotic divisions required in the plant's microsporangia and megasporangia combined?

(1) 500

(2) 1000

(3) 400

(4) 125

51) Which of the following statements regarding self-incompatibility in angiosperms is fundamentally correct according to NCERT concepts?

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(1) It is a physiological mechanism strictly dependent on the soil nutrient status.

(2) It is a genetic mechanism that prevents self-pollen from fertilizing ovules by inhibiting pollen germination or tube growth in the pistil.

(3) It promotes inbreeding depression by permanently rejecting foreign pollen.

(4) It relies on a physical, irreversible structural barrier on the stigma surface.

52) The cells of the tapetum generally have dense cytoplasm and are often multinucleate. What is the primary underlying cellular mechanism that leads to this multinucleate/polyploid condition without complete cell division?

(1) Amitosis

(2) Endomitosis and free Nuclear divisions

(3) Synchronous meiosis

(4) Repeated unequal cytokinesis

53) In a tender coconut, the surrounding white kernel and the clear fluid inside represent respectively:

(1) Cellular endosperm and Free-nuclear endosperm

(2) Free-nuclear endosperm and Cellular endosperm

(3) Perisperm and Endosperm

(4) Cotyledon and Free-nuclear endosperm

54) In dicot embryos, the cylindrical portion below the level of cotyledons is the _____ that terminates at its lower end in the radicle, while in monocots (grasses), the radicle and root cap are enclosed in an undifferentiated sheath called _____.

(1) Epicotyl; Coleoptile

(2) Hypocotyl; Coleorhiza

(3) Hypocotyl; Coleoptile

(4) Epicotyl; Coleorhiza

55) The relationship between the moth Pronuba and the Yucca plant is a classic example of obligate mutualism. Which of the following is true regarding this specific interaction?

(1) The moth feeds on the nectar while passively dusting pollen onto its wings.

(2) The moth deposits its eggs in the locule of the ovary, and the flower is pollinated by the moth in turn.

(3) The plant provides a safe site for the moth to pupate inside its hollow stem.

(4) The moth larvae aggressively feed on the stigma, accidentally transferring pollen.

56) An agricultural scientist is conducting artificial hybridization on a monoecious plant species bearing separate staminate and pistillate flowers on the same plant. Which of the following steps is inherently unnecessary for this process?

(1) Emasculation of the pistillate flower bud

(2) Bagging of the pistillate flower bud before opening

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(3) Dusting of desired pollen on the receptive stigma

(4) Re-bagging the pistillate flower after pollination

57) In a given plant species, the somatic leaf cell chromosome number is 30. What will be the total number of chromosomes in the central cell of the mature, unfertilized embryo sac?

(1) 15

(2) 30

(3) 45

(4) 60

58) Seeds of *Lupinus arcticus* (Arctic Lupine) excavated from the Arctic Tundra are renowned in biology for germinating and flowering after an estimated dormancy record of:

(1) 2000 years

(2) 10,000 years

(3) 100 years

(4) 5000 years

59) In a mature, fully dehisced anther ready to shed pollen, which of the following microsporangial wall layers are usually entirely degenerated and functionally absent?

(1) Epidermis and Endothecium

(2) Endothecium and Middle layers

(3) Middle layers and Tapetum

(4) Only Tapetum

60) The generative cell in a typical 2-celled pollen grain floats in the cytoplasm of the much larger vegetative cell. Structurally, the generative cell is _____ shaped and possesses _____ cytoplasm.

(1) Spherical; dense

(2) Spindle; dense

(3) Irregular; highly vacuolated

(4) Spindle; sparse

61) In fruits like apple, strawberry, and cashew, the fruit is botanically classified as a 'false fruit' (pseudocarp). What developmental anomaly leads to this classification?

(1) The fruit develops completely without fertilization (parthenocarpy).

(2) The ovary completely degenerates and only the thalamus forms the fruit.

(3) The Receptacle contributes significantly to fruit formation along with the ovary

(4) The fruit develops from the apocarpous pistils of a single multi-carpellary flower

62) Evaluate the following statements regarding Megasporogenesis and the development of the female gametophyte:
I. The Megaspore Mother Cell (MMC) differentiates in the micropylar region of the nucellus.

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II. Meiosis in the MMC results in a linear tetrad of four megaspores, out of which the chalazal one is usually functional.
III. The functional megaspore undergoes three consecutive meiotic divisions to form the 8- nucleate embryo sac.
IV. Six of the eight nuclei are subsequently surrounded by cell walls and organized into true cells.
Which of the given statements are correct?

(1) I, II, and IV
(2) I, II, and III
(3) II, III, and IV
(4) I, III, and IV

63) Consider the following statements regarding endosperm and embryo development in angiosperms:
I. Endosperm development invariably precedes embryo development to ensure guaranteed nutrition.
II. In seeds like castor and coconut, the endosperm is completely consumed by the developing embryo before seed maturation.
III. The primary endosperm nucleus repeatedly divides mitotically without cytokinesis to form free nuclei in the initial stages in coconut.
Which of the statements is/are INCORRECT?

(1) I and II
(2) Only II
(3) Only III
(4) II and III

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64) Match the embryonic structures in grass (monocot) seeds found in Column I with their correct anatomical descriptions in Column II:

Column I (Structure)	Column II (Description)
P. Scutellum	1. Solid, undifferentiated sheath covering the radicle
Q. Coleoptile	2. Remnant of the second cotyledon in some grasses
R. Epiblast	3. Hollow foliar structure enclosing the shoot apex
S. Coleorhiza	4. Single, large, shield-shaped cotyledon situated laterally

(1) P-4, Q-3, R-2, S-1
(2) P-4, Q-1, R-2, S-3
(3) P-2, Q-3, R-4, S-1
(4) P-4, Q-3, R-1, S-2

65) In the following question, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.
Assertion (A): In a normal chasmogamous flower, complete autogamy is virtually prevented.

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Reason (R): Complete autogamy in chasmogamous flowers requires strict synchrony in pollen release and stigma receptivity, as well as close physical proximity of anthers and stigma, which is rarely absolute in open flowers.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true and R is false.

(4) A is false and but R is true.

66) In pea plants, the allele for starch synthesis (B) shows incomplete dominance for starch grain size but complete dominance for seed shape. If a heterozygous plant (Bb) is selfpollinated, what is the probability of obtaining offspring with both round seeds and intermediate-sized starch grains?

(1) 1/4

(2) 1/2

(3) 3/4

(4) 3/16

67) A pedigree shows myotonic dystrophy (autosomal dominant) in a family. An affected heterozygous male marries an unaffected female. What is the precise probability that their first child is an affected daughter?

(1) 1/2

(2) 1/4

(3) 3/4

(4) 1/8

68) In Morgan’s experiments with Drosophila, Cross A (yellow body, white eye) showed 1.3% recombination, while Cross B (white body, miniature wing) showed 37.2% recombination. If the white eye gene is mapped between the yellow body and miniature wing genes, what is the expected map distance between the yellow body and miniature wing genes?

(1) 35.9 cM

(2) 38.5 cM

(3) 37.2 cM

(4) 98.7 cM

69) Sickle-cell trait confers resistance to malaria. The mutation involves substitution of Valine for Glutamic acid at the 6th position of the beta-globin chain. This substitution is a result of a single base change at the 6th codon of the beta-globin gene from:

(1) GAA to GUA

(2) GAG to GUG

(3) GTC to GAC

(4) CTC to CAC

70) Alpha thalassemia is controlled by two closely linked genes HBA1 and HBA2. On which human

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chromosome are these genes located, and how does the inheritance pattern differ from beta thalassemia?

(1) Chromosome 11; alpha is a quantitative defect while beta is qualitative

(2) Chromosome 16; both are quantitative, but alpha involves four alleles while beta involves two.

(3) Chromosome 16; alpha is a qualitative defect while beta is quantitative.

(4) Chromosome 11; both involve two alleles but on different loci.

71) A man with blood group A (heterozygous) marries a woman with blood group B (heterozygous). What is the probability of them having two consecutive children with blood group O?

(1) 1/4

(2) 1/8

(3) 1/16

(4) 1/2

72) During dihybrid test crosses, if genes are tightly linked on the same chromosome, the proportion of parental gene combinations is much higher than the non-parental type. Who coined the term "recombination" to describe the generation of non-parental gene combinations?

(1) Gregor Mendel

(2) Walter Sutton

(3) Thomas Hunt Morgan

(4) Alfred Sturtevant

73) Which of the following is true regarding the chromosomal basis of sex determination in domestic fowl (birds)?

(1) Females are heterogametic (ZW) and males are homogametic (ZZ)

(2) Females are homogametic (XX) and males are heterogametic (XY)

(3) Females are XO and males are XX

(4) Females are homogametic (ZZ) and males are heterogametic (ZW)

74) In a human karyotype, the presence of 47 chromosomes (44 autosomes + XXY) represents Klinefelter's syndrome. What is a key clinical feature of this condition?

(1) Short stature and furrowed tongue

(2) Webbed neck and rudimentary ovaries

(3) Overall masculine development with gynecomastia

(4) Broad palm with characteristic palm crease

75) A color-blind man marries a woman whose father was color-blind but mother was normal. What percentage of their female progeny will be color-blind?

(1) 0%

(2) 25%

(3) 50%

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(4) 100%

76) Phenylketonuria (PKU) is an autosomal recessive disorder. The affected individual lacks an enzyme that converts phenylalanine to tyrosine, leading to accumulation of:

(1) Homogentisic acid
(2) Phenylpyruvic acid and other derivatives
(3) Uric acid
(4) Melanin

77) Henking (1891) traced a specific nuclear structure all through spermatogenesis in a few insects. He observed that 50% of sperm received this structure. He named it:

(1) X chromosome
(2) Y body
(3) X body
(4) Barr body

78) In Mendelian dihybrid crosses, the probability of obtaining an offspring with the exact genotype 'AaBB' from a cross between AaBb × AaBb is:

(1) 1/16
(2) 1/8
(3) 1/4
(4) 3/16

79) Consider a pedigree where normal parents have an affected son and an affected daughter. The trait cannot be:

(1) Autosomal recessive
(2) X-linked recessive
(3) Autosomal dominant
(4) Both (2) and (3)

80) A point mutation resulting in the change of a single base pair in the gene for betaglobin chain results in sickle-cell anemia. Such a mutation where a purine is replaced by a pyrimidine (or vice versa) is specifically called a transversion. The mRNA codon corresponding to the mutant beta-globin is:

(1) GAG
(2) GUG
(3) UUG
(4) CUG

81) According to the Chromosomal Theory of Inheritance, the behavior of chromosomes is parallel to the behavior of genes. Which of the following is true for BOTH genes and chromosomes?

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(1) They both mutate at the exact same rate.

(2) They both segregate at the time of gamete formation such that only one of each pair is transmitted to a gamete.

(3) They both always show independent assortment during meiosis II.

(4) They are visible under a light microscope during interphase.

82) In grasshoppers, sex determination is of the XO type. Therefore, the total number of chromosomes in a male grasshopper is:

(1) One more than a female

(2) One less than a female

(3) Equal to a female

(4) Double that of a female

83) Which symptom is NOT correctly associated with Down's syndrome?

(1) Partially open mouth with furrowed tongue

(2) Broad palm with characteristic palm crease

(3) Physical, psychomotor, and mental development is retarded

(4) Sterility due to rudimentary gonads

84) Haemophilia is a sex-linked recessive disease. The possibility of a female becoming haemophilic is extremely rare because:

(1) Mother of such female has to be at least carrier and father should be haemophilic.

(2) Both mother and father must be homozygous recessive.

(3) It is lethal in females during embryonic development.

(4) The Y chromosome masks the effect of the haemophilia gene.

85) Sturtevant used recombination frequencies to map genes on chromosomes. If gene A and B show 5% recombination, B and C show 15%, and A and C show 20%, what is the linear order of these genes?

(1) A - C - B

(2) A - B - C

(3) B - A - C

(4) C - B - A

86) **Assertion (A):** In humans, the sex of the child is determined by the father and not by the mother.
Reason (R): Males produce two types of gametes, 50% carrying the X chromosome and 50% carrying the Y chromosome, whereas females produce only one type of ovum with an X chromosome.
Select the correct option:

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) A is true but R is false.

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(4) A is false but R is true.

87) Read the following statements carefully:
I. Mendel published his work on inheritance of characters in 1865 but it remained unrecognized till 1900.
II. One reason for Mendel's lack of recognition was that his concept of 'factors' as stable and discrete units was not accepted by his contemporaries.
III. De Vries, Correns, and von Tschermak independently rediscovered Mendel's results on the inheritance of characters.
How many of the above statements are correct?

(1) None
(2) Only one
(3) Only two
(4) All three

88) Consider the following statements regarding mutations and genetic disorders:
I. Deletion and insertions of base pairs of DNA cause frame-shift mutations.
II. Cancer cells commonly show chromosomal aberrations.
III. Colour blindness occurs in about 8% of males and only about 0.4% of females.
Which of the statements are TRUE?

(1) I and II only
(2) II and III only
(3) I and III only
(4) I, II and III

89) Identify the incorrect statements regarding pedigree analysis:
I. In a pedigree chart, a square represents a male and a circle represents a female.
II. Solid/shaded symbols represent unaffected individuals.
III. Consanguineous mating is represented by a double line between male and female symbols.
IV. A diamond symbol indicates sex unspecified.
Select the INCORRECT statement(s):

(1) II only
(2) III and IV only
(3) I and II only
(4) IV only

90) Match the genetic disorders in Column I with their chromosomal/genetic basis in Column II.

Column I (Disorder)	Column II (Basis)
P. Down's Syndrome	1. 44 Autosomes + XO
Q. Turner's Syndrome	2. Autosomal recessive mutation on Chromosome 11
R. Klinefelter's Syndrome	3. Trisomy of Chromosome 21

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S. Beta-Thalassemia	4. 44 Autosomes + XXY

Choose the correct matching:

(1) P-3, Q-1, R-4, S-2

(2) P-3, Q-4, R-1, S-2

(3) P-2, Q-1, R-4, S-3

(4) P - 1, Q - 3, R - 4, S - 2

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	4	1	1	3	4	3	1	4	4	2	2	4	1	3	1	4	1	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	1	4	1	2	2	2	4	4	4	3	1	2	4	3	2	4	3	2
Q.	41	42	43	44	45															
A.	3	3	1	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	2	3	4	2	4	4	2	4	4	2	1	3	3	2	1	2	1	1	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	1	4	3	1	3	3	2	1	3	4	2	2	4	3	4	1	4	4	4
Q.	86	87	88	89	90															
A.	1	2	4	1	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	1	2	3	3	1	4	1	3	4	3	2	2	4	1	2	4	4	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	3	1	3	1	4	3	2	4	1	4	4	1	2	3	2	1	3	2	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	3	3	2	2	2	3	2	1	2	2	1	2	2	1	2	2	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	1	2	1	4	2	2	2	2	2	3	3	1	3	3	2	3	2	1	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	4	1	2	1	4	4	1	1										

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SOLUTIONS

PHYSICS

1) $\vec{F} = I(\vec{\ell} \times \vec{B}) = 4(8\hat{i} \times 5\hat{k}) = -160\hat{j}\,N$

2)

Conceptual

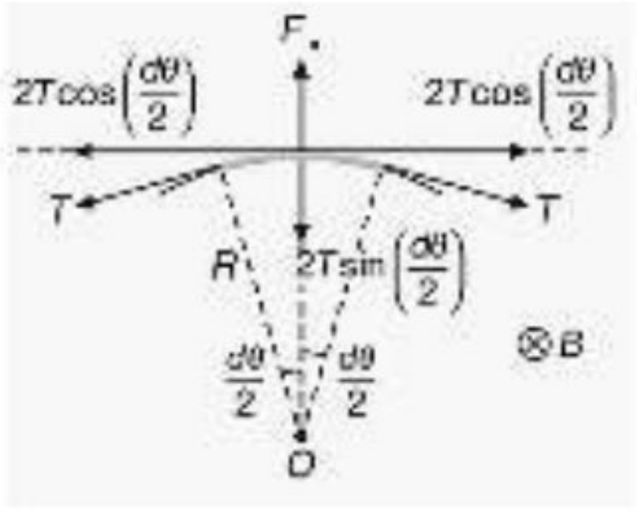
3)

Conceptual

4)

Conceptual

5)



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Magnetic force on the circular arc is Fm = BIdl.

$F_m = BI(Rd\theta)$

For the arc to be in equilibrium, $F = 2T\sin\left(\frac{d\theta}{2}\right)$

For small angle dθ, $\sin\left(\frac{d\theta}{2}\right) \approx \frac{d\theta}{2}$

$\rightarrow Td\theta = BIRd\theta$

$\Rightarrow T = BIR$

Since, $L = 2\pi R$

$\rightarrow T = BIR = \frac{BIL}{2\pi}$

6)

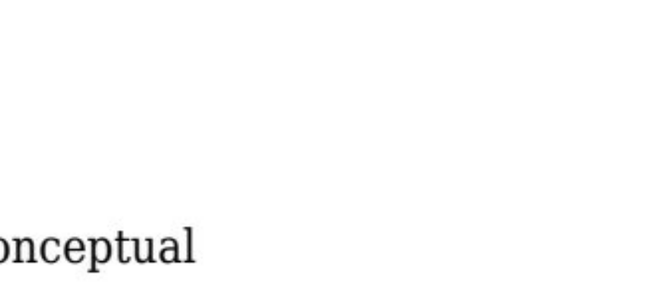
Conceptual

7)

Conceptual

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8)



Towards left of both wires direction of B is downward and at mid point between two wires, magnetic field is zero.

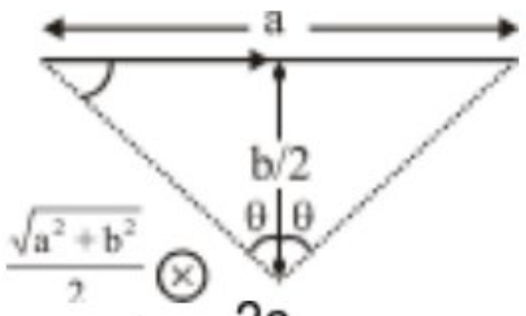
9)

Conceptual

10) $B_1 = \frac{\mu_0 i}{4\pi\, b/2}\, 2\sin\theta$

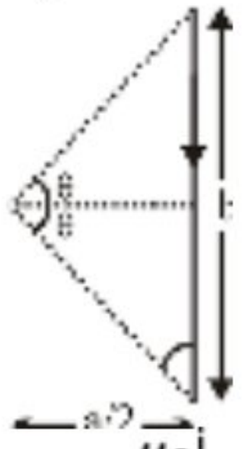
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$$= \frac{\mu_0 i}{2\pi b} \frac{2a}{\sqrt{a^2+b^2}}$$

$$B_1 = \frac{\mu_0 i}{\pi b} \frac{a}{\sqrt{a^2+b^2}}$$

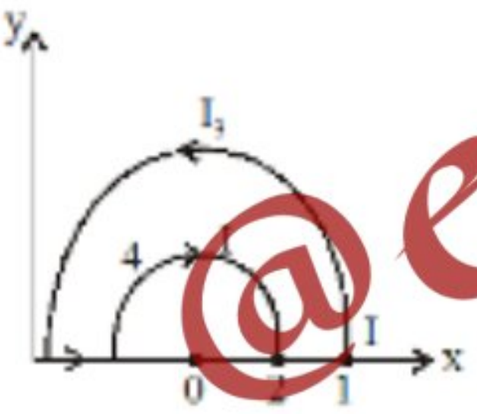


$$B_2 = \frac{\mu_0 i}{4\pi \frac{a}{2}} \frac{2b}{2\sqrt{a^2+b^2}} = \frac{\mu_0 i}{\pi a} \frac{b}{\sqrt{a^2+b^2}}$$

$$B_{net} = 2B_1 + 2B_2$$

$$= \frac{2\mu_0 i}{\pi \sqrt{a^2+b^2}} \left[\frac{a}{b} + \frac{b}{a}\right] = \frac{2\mu_0 i}{\pi ab} \sqrt{a^2+b^2}$$

11)



$$B_1 = \frac{\mu_0 I}{4\pi \times 1.5a}(-\hat{j})$$

$$B_2 = \frac{\mu_0 I}{4\pi \times 0.5a}(-\hat{j})$$

$$B_3 = \frac{\mu_0 I}{4 \times 1.5a}(+\hat{k})$$

$$B_4 = \frac{\mu_0 I}{4 \times 0.5a}(-\hat{k})$$

$$\therefore \vec{B}_{net} = \frac{\mu_0 I}{4\pi a}(2) \times \frac{4}{3}(-\hat{j}) \frac{\mu_0 I 2}{4a} \times \frac{2}{3}(-\hat{k})$$

$$\frac{\mu_0 I}{3\pi a}\sqrt{4+\pi^2}$$

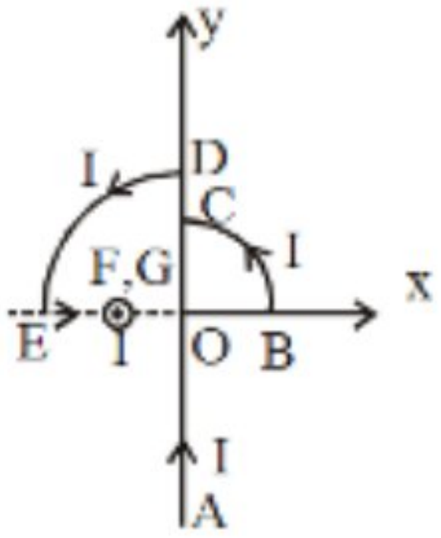
12)

Conceptual

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13)

$$\vec{B}_0 = \vec{B}_{AO} + \vec{B}_{OB} + \vec{B}_{BC} + \vec{B}_{CP} + \vec{B}_{DE} + \vec{B}_{EF} + \vec{B}_{FG}$$

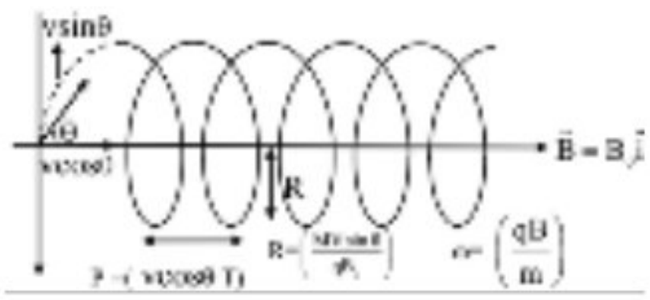
$(\because B_{AO} = B_{OB} = B_{CD} = B_{EF} = 0)$

$$\vec{B}_0 = \left(\frac{\mu_0 I}{8R} + \frac{\mu_0 I}{8 \times 2R}\right)\hat{k} + \frac{\mu_0 I}{4\pi R}\hat{j}$$

$$\frac{\mu_0 I}{8R} \times \frac{3}{2}\hat{k} + \frac{\mu_0 I}{4\pi R}\hat{j}$$

$$\boxed{\vec{B}_0 = \frac{\mu_0 I}{4R}\left[\frac{3}{4}\hat{k} + \frac{1}{\pi}\hat{j}\right]}$$

14)



$$r_1 = \frac{mv\sin\theta}{qB_0} = \frac{mv\sin 30^\circ}{qB_0}$$

$$T_1 = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$

For ptcl. $P_1 = (v\cos 30)T$

$$= (v\cos 30)\frac{2\pi m}{qB}$$

For particle 2,

$$T_2 = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$

$$r_2 = \frac{mv\sin 60^\circ}{qB_0}$$

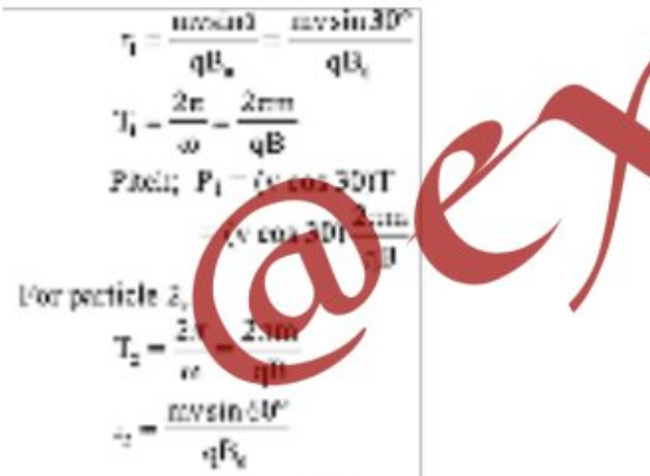
$$P_2 = (v\cos 60^\circ)\frac{2\pi m}{qB}$$

$$\Rightarrow \quad n = \frac{T_1}{T_2} = 1$$

$$\frac{r_1}{r_2} = \frac{\sin 30^\circ}{\sin 60^\circ} = \frac{1}{\sqrt{3}}$$

$$x = \frac{P_1}{P_2} = \frac{v\cos 30^\circ}{v\cos 60^\circ} = \sqrt{3}$$

15)



Magnetic field at 'P' due to wires (1) and (2) is:

$$B_1 = \frac{\mu_0 I}{2\pi(x\sin\alpha)} + \frac{\mu_0 I}{2\pi(x\sin\alpha)} = \frac{\mu_0 I}{2\pi(x\sin\alpha)}$$ (out of the paper)

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Now if a current of $\frac{2I}{\sin\alpha}$ is flowing in the third wire then the magnetic field due to the same will be:

$B_2 = \frac{\mu_0}{2\pi x}\left(\frac{2I}{\sin\alpha}\right)$, which will cancel B_1 if it is out of paper which is possible if the current $\frac{2I}{\sin\alpha}$ in the third wire is from right to left.

16)

Conceptual

17)

Ans:(4)
negative Y-direction

18)

$W_B = \Delta K + \Delta U$

$MB (\cos 90° - \cos 0°) = 0 + mg\frac{\ell}{2}$

$i\square^2 B = \frac{mg\ell}{2}$; $B = \frac{mg}{2i\ell}$

19) τ = BINA sin90º = BINA
Here B → magnetic produced by solenoid = $\mu_0 nI$

$B = \mu_0\frac{N}{\ell}I = 4\pi \times 10^{-7} \times \frac{500}{0.4} \times 3$

So torque on coil

τ = BINA

$= 4p \times 10^{-7} \times \frac{500}{0.4} \times 3 \times 0.4 \times 10 \times \pi \times (0.01)^2$

$\tau = 6 \times 10^{-6}$ N–m

20)

Conceptual

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(a) $\tau_{max} = MB = niAB = ni(l \times b)B$

21) $\tau_{max} = 600 \times 10^{-5} \times 5 \times 10^{-2} \times 12 \times 10^{-2} \times 0.10 = 3.6 \times 10^{-6}$ N-m.

22)

Conceptual

23)

When similar poles are on same side time period of oscillation T_1 is given by

$$T_2 = 2\pi\sqrt{\frac{I_1 + I_2}{(m_1 + m_2)B_H}}$$ [given]

$m_1 = 2m, m_2 = m$

$$T_2 = 2\pi\sqrt{\frac{I_1 + I_2}{(3m)B_H}} \quad \ldots\ldots\ldots \text{(i)}$$

When the polarity of magnet is reversed, time period of oscillation T_2 is given by

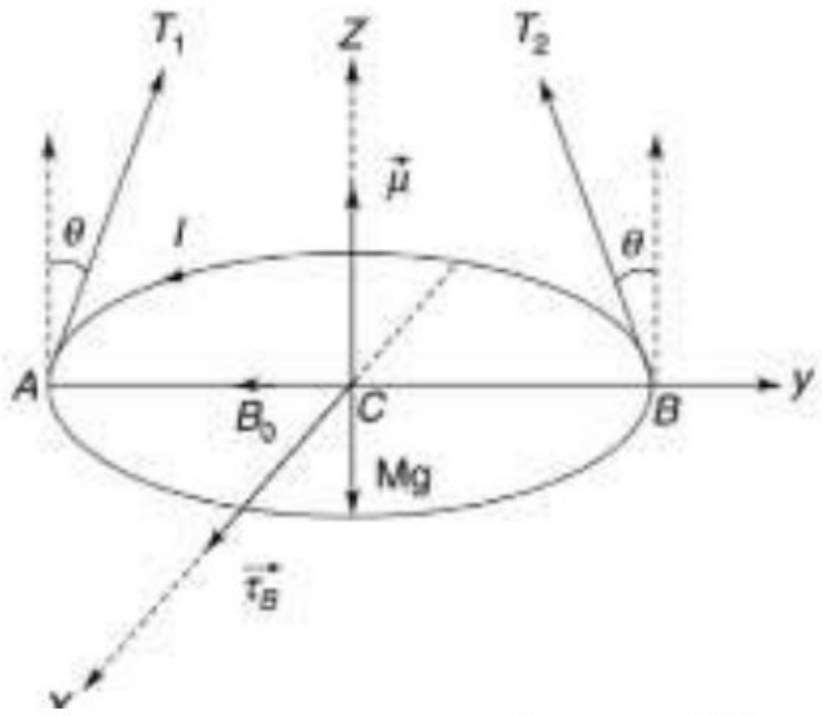
$$T_2 = 2\pi\sqrt{\frac{I_1 + I_2}{(m_1 - m_2)B_H}}$$

$$T_2 = 2\pi\sqrt{\frac{I_1 + I_2}{mB_H}} \quad \ldots\ldots\ldots \text{(ii)}$$

On comparing Eqs. (i) and (ii), we get

$T_2 > T_1$

or $T_1 < T_2$.



24)

Magnetic dipole moment $\vec{\mu} = I.\pi R^2 \hat{k}$

Magnetic torque $\vec{\tau}_B = \vec{\mu} \times \vec{B}$

$$= I\pi R^2 B_0 \left(\hat{k} \times \left(-\hat{j}\right)\right) = I\pi R^2 B_0 (\hat{i})$$

To counterbalance this torque we must have $T_1 > T_2$

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Torque about centre C

$(-T_1 \cos\theta.R + T_2 \cos\theta R)\hat{i} + I\pi R^2 B_0 \hat{i} = 0$

$\therefore \quad T_1 - T_2 = \frac{I\pi RB_0}{\cos\theta}$ ...(i)

And $(T_1 + T_2)\cos\theta = Mg$

$\Rightarrow \quad T_1 + T_2 = \frac{Mg}{\cos\theta}$ ....(ii)

Solving (i) and (ii)

$T_1 = \frac{1}{2\cos\theta}(Mg + \pi IRB_0)$ and $T_2 = Mg - \pi IRB_0$

25)

Conceptual

26) $W_1 = MB(\cos 0° - \cos 90°) = MB$

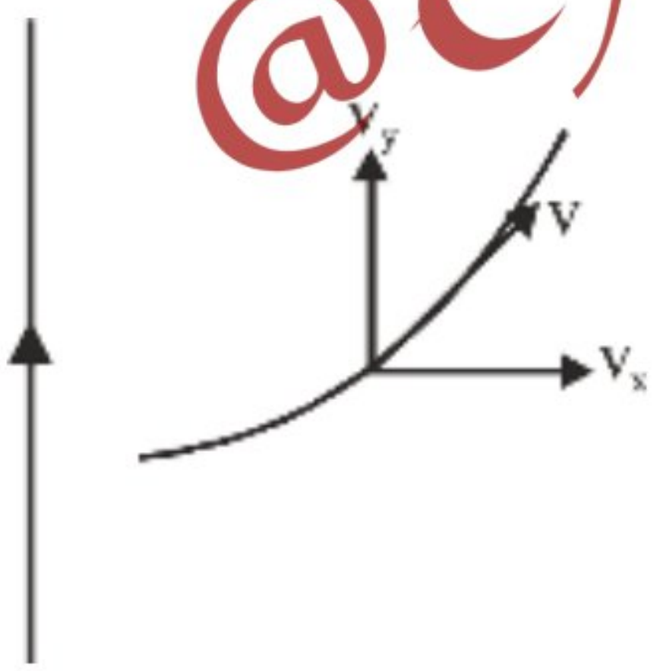
$W_2 = MB(\cos 0° - \cos 60°) = \frac{MB}{2}$

$W_1 = nW_2$

$MB = n\frac{MB}{2}$

$n = 2$

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27)

By impulse momentum theoram

$f_y = qv_xB$

$\int qv_xB\,dt = mv_f - 0$ .....(i)

$\frac{dx}{dt} = v_x \Rightarrow dx = v_xdt$

and $\frac{\mu_0 i}{2\pi x}$

$\int_r^{r_f} \frac{q\mu_0 i}{2\pi x}dx = mv_f$

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$$\frac{q\mu_0 i}{2\pi}\ell n\left(\frac{r_f}{r_i}\right) = mv$$ [As magnetic field do no work $v_f = v$]

$r_f = e^{\frac{2\pi mv}{q\mu_0 i}}$

28)

$W_{electrical} + W_{magnetic} = \Delta K$

$\Rightarrow qE^0 . X^0_{25} = \frac{1}{2}m(5)^2 = 0$

$y_0 = \frac{}{2\alpha E_0}$

29)

Conceptual

30) $\vec{E}$ is parallel to $\vec{B}.\vec{v}_0$ is perpendicular to both. Therefore path of the particle is a helix with increasing pitch. Speed of particle at any time t is $v = \sqrt{v_x^2 + v_y^2 + v_z^2}$ ....(i)

Here, $v_y^2 + v_z^2 = v_0^2$

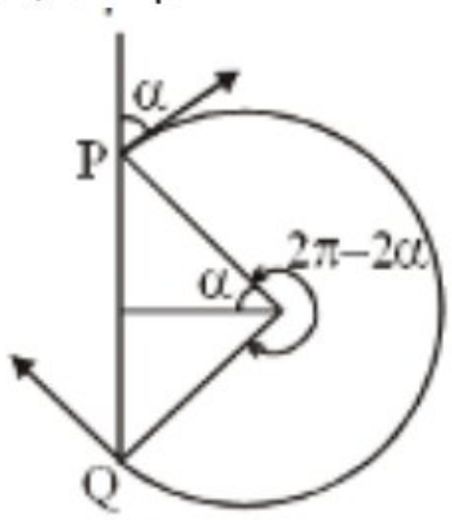
$v = 2v_0$

putting this in equation (i)

$v_x = \sqrt{3}v$ and $v_x = \frac{qE_0}{m}t$

$t = \frac{\sqrt{3}mv_0}{qE_0}$

31) $\alpha = \beta$



$t = T\left\{\frac{2\pi - 2\alpha}{2\pi}\right\}$

$PQ = 2R \sin \alpha$

32)

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NOVA

Conceptual

33)

Conceptual

34)

Conceptual

35)

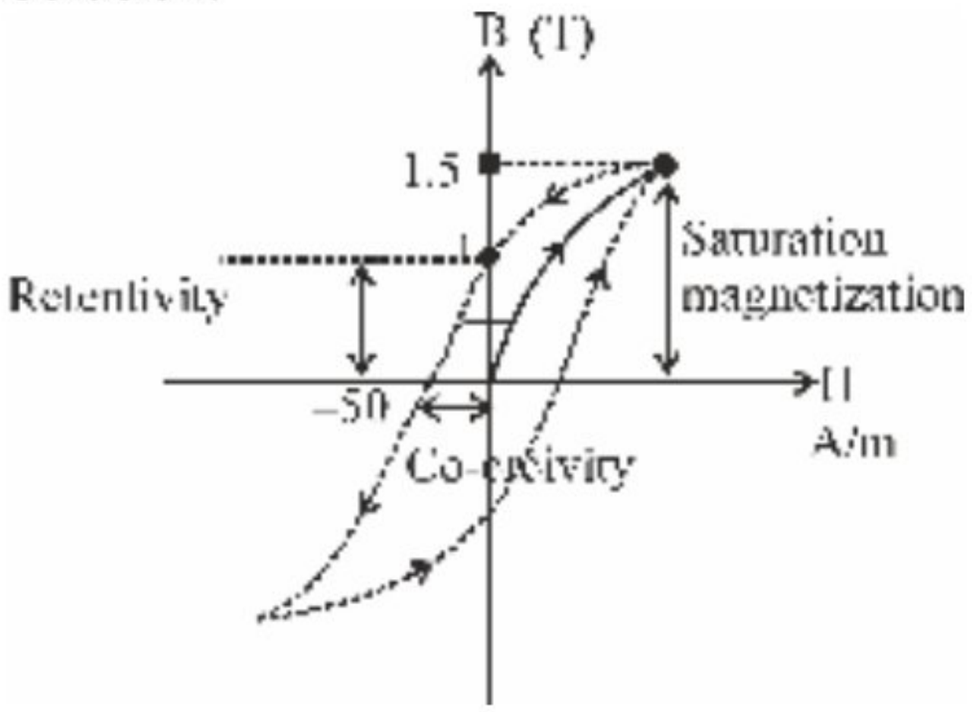
Conceptual

36)

Conceptual

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37) **Question Explanation:** The question asks you to identify three properties—retentivity, coercivity, and saturation—from a B-H (magnetic induction vs. magnetizing field) curve for a ferromagnetic material.
Concept: B-H Hysteresis Loop
Solution:



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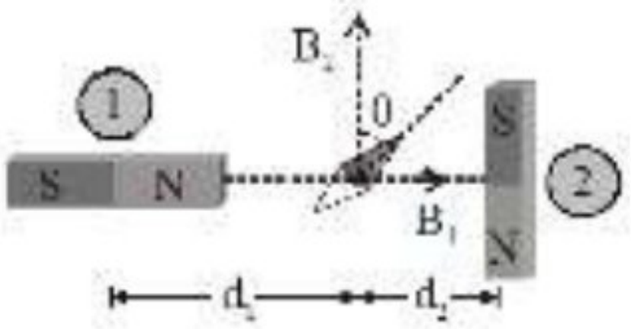
Retentivity = 1.0 T
Co-ercivity = 50 A/m
Saturation = 1.5 T
Final answer : Option (2)
1.0 T, 50 A/m and 1.5 T

38)

Conceptual

39)

In equilibrium $B_1 = B_2 \tan\theta$



$$\Rightarrow \frac{\mu_0}{4\pi}.\frac{2M}{d_1^3} = \frac{\mu_0}{4\pi}.\frac{M}{d_2^3}\tan\theta$$

$$\Rightarrow \frac{d_1}{d_2} = (2\cot\theta)^{1/3}$$

40)

ConceptualConceptual

41)

Conceptual

42) Initially magnetic moment of system

$M_1 = \sqrt{M^2 + M^2} = 2M$ and moment of inertia

$I_1 = I + I = 2I$

Finally when one of the magnet is removed then

$M_2 = M$ and $I_2 = I$

So $T = 2\pi\sqrt{\frac{I}{MB_H}}$

$$\frac{T_1}{T_2} = \sqrt{\frac{I_1}{I_2}\times\frac{M_2}{M_1}} = \sqrt{\frac{2I}{I}\times\frac{M}{\sqrt{2}M}}$$

$$\Rightarrow T_2 = \frac{2^{5/4}}{2^{1/4}} = 2\,\text{sec}.$$

43) Flux through the ring is $\phi = B.A = \left(2\hat{j} + t^2k\right).\left(\pi R^2\hat{k}\right) = \pi R^2 t^2$

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$\therefore$ Induced emf $|\varepsilon| = \frac{d\phi}{dt} = 2\pi R^2 t$

If induced electric field is E_{in} then

$E_{in}\ 2\pi R = \varepsilon$

$E_{in}\ 2\pi R = 2\pi R^2 t$

$E_{in} = Rt$ ...(i)

Current in the ring is $i = \frac{\varepsilon}{r} = \frac{2\pi R^2 t}{r}$ [Clock wire]

Magnetic dipole moment $\vec{\mu} = \left(\pi R^2 i\right)\left(-\hat{k}\right) = \frac{2\pi^2 R^4 t}{r}\left(-\hat{k}\right)$

Torque on the ring $\vec{\tau} = \vec{\mu} \times \vec{B} = \frac{4\pi^2 R^4 t}{r}$(i)

The ring will begin to topple when magnetic torque exceeds the torque due to weight about an axis tangential to the ring and parallel to x axis.

In the figure the weight (Wof the ring is acting into the plane of the figure.

$\Rightarrow \quad t_0 = \frac{mg.r}{4\pi^2 R^3} = \frac{\pi \times 10 \times 4}{4\pi^2 \times \left(\frac{1}{2}\right)^3} = 20\ \text{sec.}$

Induced field at t_0 from equation (i) is

$Ein = \frac{mgr}{4\pi^2 R^2}\frac{\pi \times 10 \times \pi}{4\pi^2\left(\frac{1}{2}\right)^2} = 10\ \text{V/m}$

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44) Conceptual

45) Conceptual

CHEMISTRY

46)

Answer : 4

Concept:

The reaction between mercuric iodide (Hgl_4^{2-})and potassium iodide (Kl) involves the formation of a stable complex ion. Here's the reaction

Explanation: The reaction between **mercury (II) iodide (Hgl_2)** and **potassium iodide (Kl)** is notable because they interact to form a soluble complex. This is utilized in the preparation of **Nesser's reagent**,

Reaction Steps

1. Formation of the Complex:

$Hgl_2 + 2Kl \rightarrow K_2[Hgl_4]$

47) **Explain Question:**

Question is asking about the compound When a mixture of solid NaCl, solid $K_2Cr_2O_7$ is heated with concentrated H_2SO_4, orange red vapours are obtained

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Concept
This question is based on Colors of vapours), $K_2Cr_2O_7$ properties

Solution:
Chromyl chloride test $K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \rightarrow 2CrO_2Cl_{2(g)} + 4NaHSO_4 + 3H_2O$
CrO_2Cl_2 is chromyl chloride and it shows orange-red vapours.

Final Answer: Option (2)

48) **Question Explanation**
Question is asking about an aqueous solution of gas (X) that turns red litmus paper into blue.
Concept
The concept based on Litmus paper , Acidic and basic nature.
Explanation
The gas that turns red litmus paper blue when dissolved in water is ammonia (NH_3). Ammonia is a basic gas, and when dissolved in water, it forms ammonium hydroxide (NH_4OH), which is a weak base and turns red litmus paper blue.
Final Answer (3)

49)

Correct Option is: (4)

50) **Question Explanation:**
Question is asking that Dil. HCl + H_2S is the group reagent of which group

Concept:
This question is based on Group reagent

Solution:
Dil. HCl + H_2O is the group reagent of IIndgroup

Hence,
Correct Option:2

51) **Question Explanation:**
Question is asking about the sulphide which gives white ppt. When H_2S is passed through ammoniacal solution.

Concept:
This question is based on Precipitate, solubility

Solution:
In an ammoniacal solution (NH_4OH + NH_4Cl), H_2S reacts with zinc ions (Zn^{2+}) to give:
$Zn^{+2} + H_2S \rightarrow ZnS\downarrow + 2H+$
Zinc sulfide (AnS) is white precipitate.

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NOVA

Hence,
Correct Option: 4

52) **Quesiton Explanation:**
Question is asking about the compound does not dissolve in concentrated ammonium hydroxide solution.

Concept:
This question is based on Solubility, General reactions

Solution:
AgI (silver iodide) does NOT dissolve in concentrated ammonium hydroxide (NH_4OH) solution.
$AgCl\ (s)\ 2NH_3 \rightarrow [Ag(NH_3)_2]^+ + Cl^-$
AgI, however, is much less soluble and does not form this complex easily due to its very low solubility and stronger lattice energy.
Therefore, AgI remains as a yellow precipitate and does not dissolve in concentrated ammonium hydroxide.

Hence,
Correct Option: 4

53)

Conceptual

54) **Question Explanation:**
Identification of ion which from violet layer in Chloroform when its salt is added with chlo water

Concept:
Layer test for the detection of halide ions

Solution:
Cl_2 water acts as oxidising agene
$Cl_{2+}\ 2I^- \rightarrow\ 2Cl^- +\ I_2^-$
(I_2) is soluble in chloroform due to organic solvent
when Mixture of chloroform and I_2 is shaken , I_2 dissolves and form a violet/ purple colour layer

Final answer: Option (4)

55)

Conceptual

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56)

Question Explanation :
Question is asking abou the colour of $CoO.Al_2O_3$

Concept :
This question is based on Colours of salts.

Solution :

$CoO.Al_2O_3$ (cobalt aluminate) also known as cobalt blue has blue colour. It is formed by (CoO) and Al_2O_3 combination.

Hence,
Option (2) is correct : Blue

57) **Explain Question:** Identify the ion detected by using Nessler's reagent.

Concept: Qualitative analysis

Solution: Nessler's reagent reacts specifically with ammonium ions to give a brown coloration or
precipitate of iodide of Millon's base, confirming the presence of NH_4^+ in solution.

Final Answer: Option (1)

58)

Conceptual

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59)

Question Explanation:
We are asked which metal does not give the borax bead test

Concept:
Qualitiative analysis (Borax bead test)

Solution:
• The borax bead test is used for transition metals (like Cr, Ni, Mn, Co, Cu) which form colored metaborates.
• Na is an alkali metal, not a transition metal and does not form characteristics coloured borates in the bead test.
• Therefore, sodium does not give the borax bead test.

Final answer: Option (3)

60)

Question Explanation:

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We are asked which compound is formed when ferric ion produces a Prussian blue coloured precipitate.

Concept:
Qualitative analysis (Prussian blue test for Fe^{3+})
Solution:
• Ferric ions (Fe^{3+}) react with potassium ferrocyanide $Fe_4[(CN)_6]_3$.
• This compound is known as Prussian blue, which gives the characteristic deep the precipitate.

Final answer: Option (2)

61)

Explanation :- Question is asking about the greenish yellow gas liberated when MnO2 and H2SO4 added to NaCl,

Concept – This question is based on Identification and colour of gas

Solution The greenish-yellow gas liberated when MnO_2 (manganese dioxide) and H_2SO_4 (sulfuric acid) are added to NaCl (sodium chloride) is chlorine gas (Cl_2) as

$MnO_2+4NaCl+4H_2SO4 \rightarrow MnSO_4+4NaHSO_4+2H_2O+Cl_2$

Final Answer 1

62) **Explain Question:**
Identifying property not essential for catalysts asking about catalyst features.

Concept:
Catalysts need vacant orbitals, surface valency, and variable oxidation states.

Solution:
Catalysts commonly work via variable oxidation states, vacant orbitals, and valencies on the surface. High ionisation energy is not required and does not help catalysis.

Final Answer: Option (2)

63) **A. Question Explanation**

Identifying the product formed when $KMnO_4$ reacts with conc. H_2SO_4 and its decomposition.

B. Concept

Concentrated acids convert permanganate into unstable higher oxides like Mn_2O_7.

D. Solution

When $KMnO_4$ is treated with concentrated H_2SO_4, it forms dark green oily Mn_2O_7, an explosive

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oxide of manganese. Upon heating, Mn_2O_7 decomposes to form MnO_2. Thus, the pair of compounds produced are Mn_2O_7 and MnO_2 respectively.

E. Final Answer OPTION 1

64)

A. Question Explanation

Identifying the green coloured powder blown in air when (NH4)2Cr2O7 is heated.

B. Concept

Thermal Decomposition of Ammonium Dichromate.

D. Solution

(NH4)2Cr2O7 decomposes to produce N2, H2O, and green Chromium(III) Oxide (Cr2O3) powder.

$$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + H_2O$$

E. Final Answer OPTION 1

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65) **Explain Question:** Identifying elements stabilizing highest oxidation sates asking about strong electronegative partners.

Concept: Oxidation states

Solution: Highest oxidation states of transition elements are stabilized by highly electronegative and small atoms forming strong bonds. Oxygen and fluorine best stabilize high oxidation states due to strong M–O and M–F bonding.

Final Answer: Option (1)

66) **Question Explanation :** Identify which metals exist in volatile or easily vaporisable form.

Concept : Volatility increases for metals with low boiling points and weak metallic bonding.

Solution: Among the given options, Zn, Cd, and Hg belong to group 12 and have low melting and boiling points due to completely filled d^{10} configuration causing weak metallic bonding. Hence, they vaporize easily and are called volatile metals. Other options contain high-melting transitions or nonmetals.

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Final Answer : OPTION 2

67)

Question Explanation : Identify the coloured compound formed when MnO_2 is fused with KOH and O_2.

Concept : Fusion with alkali and O_2 oxidizes Mn^{4+} to Mn^{6+} (manganate).

Solution :

$MnO_2 + 2KOH + O_2 \rightarrow K_2MnO_4$ (green compound).

Final Answer : Option (1)

68)

Conceptual

69) NCERT Pg. # 219

70)

Conceptual

71)

Conceptual

72)

Question Explanation: We evaluate stability of oxidation states and electronic configurations in lanthanides.

Concept: General properties of F block elements

Solution:

Lanthanides mainly show +3 oxidation state, with extra stability for half-filled and fully filled f-configurations.

(1) Eu^{2+} (f^7), Yb^{2+} (f^{14}) → extra stability → Correct

(2) Ce^{4+} is stable due to f^0 configuration → Correct

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(3) Lanthanides mainly show **+3**, not +2 → Incorrect

(4) Gd^{3+} = f^7 (half-filled) → extra stability → Correct

Final Answer: Option (3)

73)

ConceptualConceptual

74) **Question Explanation**: CrO_4^{2-} to $Cr_2O_7^{2-}$ equilibrium and pH.
Concept: Chromate-Dichromate Equilibrium.
Solution: The equilibrium between **chromate (yellow)** and **dichromate (orange)** is $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$.

A. **$CrO_4^{2-} \rightarrow Cr_2O_7^{2-}$ (Yellow to Orange)**: Occurs by **adding H+**, meaning pH<7 (acidic medium). Hence, x=6 is a common value.
B. **$Cr_2O_7^{2-} \rightarrow CrO_4^{2-}$ (Orange to Yellow)**: Occurs by **removing H+** (adding OH^-), meaning pH>7 (basic medium). Hence, y=8 is a common value. **Final Answer**: 6, 8

$$\underset{\text{(yellow)}}{CrO_4^{2-}} \underset{\text{pH>7, Basic}}{\overset{\text{pH<7, Acidic}}{\rightleftharpoons}} \underset{\text{(orange)}}{Cr_2O_7^{2-}}$$

75)

Conceptual

76)

Conceptual

77)

Question Explanation :

We must determine the correct ground-state electronic configuration of thorium (Th, Z = 90).

Concept :

Actinide elements fill 5f, 6d, and 7s orbitals; Th begins 5f involvement.

Solution :

For Th (Z = 90): the configuration is $5f^0\ 6d^2\ 7s^2$.

The 5f orbital is nearly empty, and two electrons occupy 6d with two in 7s.

Final Answer : Option (2)

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78)

Conceptual

79)

Question Explanation :

The question asks us to calculate the number of Faradays required to reduce a given amount of Ni^{2+} to Ni metal.

Concept :

A. Gram Equivalent
B. Faraday's Laws of Electrolysis

Solution :
Given:
5 gram equivalents of Ni^{2+}

Since 1 gram equivalent of any substance requires 1 Faraday of charge for reduction (or oxidation), 5 gram equivalents of Ni^{2+} will require 5 Faradays.

Final Answer : Option (4)

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80)

Question Explanation
Determine which substance can oxidize bromide ions (Br^-) to bromine gas (Br_2).
Concept
A stronger oxidizing agent than Br_2 can liberate Br_2 from KBr.
Solution
$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$ (spontaneous since $E°(Cl_2/Cl^-) > E°(Br_2/Br^-)$).
Final Answer_3

81)

Question Explanation: Find products at **anode** and **cathode** during electrolysis of aqueous NaCl using **Hg cathode**.

Concept: Electrolysis

Solution:
In aqueous NaCl:
• At anode, Cl^- is preferentially oxidized to Cl_2.
• At Hg cathode, Na^+ is reduced to Na and forms Na–Hg amalgam (not H_2).
Anode reaction: $2Cl^- \rightarrow Cl_2 + 2e^-$
Cathode reaction: $Na^+ + e^- \rightarrow Na$ (dissolves in Hg as amalgam)

Final Answer: Option 4

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82)

Question Explanation: Compare the conductivities of given solutions based on ionization strength and ion concentration.
Concept: This question is based on conductivity depends on the number of ions and their mobility. Strong electrolytes ionize completely, while weak electrolytes ionize partially.
Solution:
HCl → strong acid → completely ionized → maximum ions.
NaCl → strong electrolyte → completely ionized but less mobile ions than H^+.
CH_3COOH → weak acid → partially ionized.
H_2O → very poor conductor (almost non-ionized).
Therefore, HCl gives maximum conductivity due to complete ionization and high mobility of H^+ ions.
Final Answer: Option 1

83) **Explain Question:**
To identify the function of salt bridge.

Concept:
Function of salt bridge.

Solution:
The function of salt bridge.
(i) It connect two half cells.
(ii) It maintain electrical neutrality.
(iii) It minimise liquid-liquid junction potential.
(iv) It complete internal circuit.

Final answer: Option (4)

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84)

Question Explanation: Identify the correct equation representing the standard reduction potential of zinc.
Concept: This question is based on Standard reduction potential (E°) is defined for a reduction half-cell reaction under standard conditions (1 M, 1 atm, 25°C) in aqueous solution.
Solution: For zinc, the reduction process involves Zn^{2+} ions in aqueous solution gaining two electrons to form solid zinc metal. **$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$**
Final Answer: Option 4

85)

Conceptual

86)

Question Explanation: Identifying which metal's oxide cannot be reduced by aluminium in a thermite reaction.
Concept: This question is based on Al reduces oxides of metals less reactive than itself (below

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Al in reactivity series).
Solution:
K is more reactive than Al; hence Al cannot reduce K_2O to K metal.
Final Answer: Option 1

87)

Conceptual

88) **Explanation the Question:**
The question asks to identify which coordination compound among the given options will exhibit the maximum conductivity in solution.
Concept:
This question is based on
Conductivity of Coordination Compounds
Solution:
1. $[Cr(NH_3)_3Cl_3]$: This is a neutral complex and does not dissociate into ions.
• Number of ions = 0
2. $[Cr(NH_3)_4Cl_2:]CI$: This complex dissociates into one complex cation and one chloride anion.
• $[Cr(NH_3)_4CI_2]CI \rightarrow [Cr(NH_3)_4Cl_2]^+ + Cl^-$
• Number of ions = 2
3. $[Cr(NH_3)_5CI]CI_2 \rightarrow$ This complex dissociates into one complex cation and two chloride anions.
• $[Cr(NH_3)_5CI]Cl_2 \rightarrow [Cr(NH_3)_5CI]^{2+} + 2CI^-$
• Number of ions = 3
4. $[Cr(NH_3)_6]\ Cl_3$: This complex dissociates into one complex cation and three chloride anions.
• $[Cr(NH_3)_6]Cl_3 \rightarrow [Cr(NH_3)_6]^{3+} + 3CI^-$
• Number of ions = 4

89) **Question Explanation :**
Match the column-I with column-II.

Concept :
This question is based on Theory of electrolysis.

Solution :
(i) Very dilute solution of NaCl ⇒ O_2 evolved at anode and H_2 evolved at cathode.

(ii) Very dilute solution of HCl ⇒ O_2 evolved at anode and H_2 evolved at cathode.

(iii) Concentrated solution of NaCl ⇒ H_2 evolved at cathode and Cl_2 evolved anode.

(iv) Fairly concentrated solution of NaCl ⇒ O_2 evolved at anode and Ag deposited at cathode.

Hence,

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Option (1) is correct : A → (p,q), B → (p,q), C → (q,r), D → (p,s)

90) **Explain Question:**
What is efficiency of fuel cell.

Concept:
Fuel cell.

Solution:

The efficiency of fuel cell $\eta = \frac{|\Delta G|}{|\Delta H|}$

Final answer: Option (3)

BIOLOGY

92)

Conceptual

93) **Solution :** Antigens: foreign molecules that trigger an immune response.
Antibodies: proteins produced by plasma cells that specifically recognize and neutralize antigens.
Final Answer : (1)

94)

Conceptual

95)

Question explanation: This question asks about the plant shown in the image and its product's properties.

Concept: Drug plants and their products

Solution: The plant shown in the image is Papaver somniferum (opium poppy), not Datura.

A. Heroin is derived from this plant.
B. It is a white, odorless, bitter, crystalline compound.Chemically, heroin is diacetyl morphine.
C. Thus, the correct options describing this plant and product are: c, d, f, g.

Final answer: Option (3)

96)

Question explanation: Fill in the blanks

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Concept: Common human disease

Solution:
(A)Mary Mallon___was a typhoid carrier who continued to spread typhoid for several years.

(B)Haemophilus influenzae___ is responsible for the disease pneumonia in humans.

(C)Ringworms___ is one of the most common infectious diseases in man.

(D) Pathogens that enter the ___(D)Gut___ must know a way of surviving in the stomach at low pH.
Final answer:3

97)

Conceptual

98)

NCERT-XII, Pg. # 135

104) **Question Explanation**: The question asks to identify which listed conditions are autoimmune disorders.

Concept: The question is based on autoimmunie disorders

Solution:

A. A. Myasthenia gravis → Autoimmune disorder (antibodies against acetylcholine receptors).
B. B. Rheumatoid arthritis → Autoimmune disorder (immune attack on synovial joints).
C. C. Gout → Not autoimmune; it is a metabolic disorder due to uric acid crystal deposition.
D. D. Muscular dystrophy → Genetic disorder, not autoimmune.
E. E. Systemic Lupus Erythematosus (SLE) → Autoimmune disease where the body's immune system attacks its own healthy tissues and organs, causing inflammation and damage throughout the body

Correct Set: A, B & E

Final Answer: Option 2

105)

ConceptualConceptual

109)

Question Explanation : Identify the correct statements among the given options.

Concept : The question is based on human health and diseases

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Solution : The correct answer is option 4.

A. (a) Typhoid is a vector borne disease — Incorrect; it is a water/food-borne disease caused by *Salmonella typhi*.

B. (b) T-helper cells are called as HIV factory — Incorrect; HIV infects T-helper cells, but they are not termed as “HIV factory” .Macrophages are considered as HIV factory

C. (c) Primary immune response is of low intensity and slow — Correct; it takes longer time and produces fewer antibodies.

D. (d) Hepatitis B and AIDS are example of sexually transmitted diseases — Correct; both can be transmitted sexually.

E. (e) Down syndrome is the most common sex trisomy — Incorrect; it is autosomal trisomy (trisomy 21), not sex chromosome trisomy.

F. (f) Characters of albinism also found in Alkaptonuria & Phenylketonuria — Incorrect; these are metabolic disorders with different symptoms.

Final answer : Option 4

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111)

Conceptual

112)

Conceptual

113)

Conceptual

115) **Explanation:**
(i) Myasthenia Gravis: This disorder involves autoantibodies targeting **acetylcholine receptors** at neuromuscular junctions, causing muscle weakness.
(ii) Pernicious Anemia: This condition results from autoantibodies against **intrinsic factor**, a protein needed for vitamin B12 absorption.
(iii) IDDM (Insulin-Dependent Diabetes Mellitus) / Type 1 Diabetes: This is an autoimmune attack on the **beta cells of the pancreas**, which produce insulin.
(iv) Multiple Sclerosis: This disease involves autoantibodies attacking the **myelin sheath** that surrounds and protects nerve fibers in the central nervous system.

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Correct Answer : 1. (i)-d, (ii)-c, (iii)-b, (iv)-a

117) NCERT (E) Pg. # 161

118) NCERT Pg.#149

119)

Conceptual

120)

Conceptual

121)

Conceptual

122)

Conceptual

123)

Conceptual

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126)

NCERT-XII, Pg. # 138

127)

Question Explanation: Determine the requirements for an **autograft** (transplant within the same individual).

Concept: The question is based on organ transplantation

Solution: The correct answer is option 1.

A. In autograft, tissue is transplanted within the same individual.
B. There is no risk of immune rejection, so blood group matching, MHC matching, or immunosuppressive drugs are not required.
C. Vaccination is unrelated to autograft success.

Number of requirements: Zero

Final answer: Option 1

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129)

Question explanation: Identify drugs that **reduce allergy symptoms**.

Concept: Allergy

Solution: The correct answer is **Option 2**. i, ii and iii

A. **Antihistaminic drugs (i)** → Quickly relieve symptoms by blocking histamine

B. **Adrenaline (ii)** → Acts fast in severe allergic reactions (anaphylaxis)

C. **Steroids (iii)** → Reduce inflammation and suppress the immune response, helping with symptoms like congestion and swelling.

D. **α-interferon (iv)** → Used for viral infections, **not for allergies**

Final answer: Option 4

131)

ConceptualConceptual

132)

Conceptual

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134) NCERT XII Pg. # 146, 147, 148, 149

135) a. Intravenous drug abusers are at a high risk of acquiring HIV infection – Correct. Sharing contaminated needles is a common route for HIV transmission.
b. Chemical carcinogens like UV rays and X-rays cause neoplastic transformation – Incorrect. UV rays and X-rays are physical carcinogens, not chemical carcinogens.
c. Cancer is the most dreaded infectious disease in humans – Incorrect. Cancer is not an infectious disease; it's a non-communicable disease.
d. Drugs used in chemotherapy besides targeting cancer cells, also inhibit the proliferation of normal cells in the body leading to hair loss and anaemia – Correct. Chemotherapy drugs can affect normal rapidly dividing cells, leading to side effects like hair loss and anaemia.
Correct statements: a and d (2 statements).

136) Explanation: The female plant is 6n, meaning the haploid gametophytic cells (egg, polar nuclei) are 3n. The central cell has two polar nuclei (3n + 3n = 6n). The male plant is 2n, so the male gamete is n. Zygote = Egg (3n) + Male Gamete (n) = 4n. PEN = Central Cell (6n) + Male Gamete (n) = 7n.

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137) Explanation: Microspores are arranged in a cluster of four cells (microspore tetrad) held together by a callose wall. The tapetum secretes the enzyme callase to degrade this wall, allowing the microspores to separate into individual pollen grains.

138) Explanation: Adventive polyembryony is a form of apomixis where maternal somatic cells (like the nucellus or integument) directly develop into embryos mitotically. Since they do not undergo meiosis or fertilization, they are genetically identical clones of the mother plant (AaBbCc).

139) Explanation: Guided by the filiform apparatus, the pollen tube enters one of the synergids. The synergid begins to degenerate, and the pollen tube releases its two male gametes into the cytoplasm of this degenerating synergid before they move towards the egg and central cell.

140) Explanation: 400 seeds require 400 eggs and 400 male gametes. For eggs: 1 Megaspore Mother Cell (MMC) undergoes 1 meiosis to produce 1 functional megaspore (egg). So, 400 eggs = 400 divisions. For pollen: 1 Microspore Mother Cell undergoes 1 meiosis to produce 4 pollen grains. So, 400 pollen = 400 / 4 = 100 divisions. Total meiotic divisions = 400 + 100 = 500.

141) Explanation: Self-incompatibility is an outbreeding device. NCERT clearly defines it as a genetic mechanism that prevents self-pollen (from the same flower or other flowers of the same plant) from fertilizing the ovules by actively inhibiting pollen germination or pollen tube growth in the pistil.

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142) Explanation: Tapetal cells become multinucleate or polyploid due to karyokinesis not being followed by cytokinesis (free nuclear division) or through endomitosis, which increases their metabolic capacity to nourish the developing pollen.

143) Explanation: The coconut water from a tender coconut is free-nuclear endosperm (made up of thousands of nuclei). As development proceeds, cell wall formation occurs from the periphery towards the center, turning it into the white cellular endosperm (kernel).

144) Explanation: The region below the cotyledons in a dicot is the hypocotyl. In monocots, the radicle and root cap are protected by a solid, undifferentiated sheath called the coleorhiza. (Coleoptile covers the plumule).

145) Explanation: The moth cannot complete its life cycle without the plant and vice versa. The moth deposits its eggs in the ovarian locule and simultaneously pollinates the flower. The larvae come out of the eggs as the seeds start developing, feeding on a few of them.

146) Explanation: Emasculation is the removal of anthers from a bisexual flower to prevent selfpollination. Since the pistillate flower is unisexual (has no anthers), emasculation is not required. Bagging is still needed to prevent unwanted cross-pollination.

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147) Explanation: The somatic cell (2n) = 30, meaning the haploid number (n) = 15. A mature, unfertilized central cell contains two polar nuclei, each being haploid (n+n). Therefore, the total chromosome number in the central cell is 15 + 15 = 30.

148) Explanation: The oldest record of seed viability mentioned in NCERT is for *Lupinus arcticus*, which germinated after an estimated 10,000 years of dormancy. (The 2000-year record belongs to the Date Palm, *Phoenix dactylifera*).

149) Explanation: As the anther matures and dehydrates, the tapetum is completely consumed while nourishing the pollen, and the thin-walled middle layers are crushed and degenerate. Only the epidermis and the specialized endothecium persist to aid in dehiscence.

150) Explanation: According to NCERT, the generative cell is small and floats in the cytoplasm of the vegetative cell. It is spindle-shaped with dense cytoplasm and a prominent nucleus.

151) Explanation: True fruits develop solely from the ovary. In false fruits, other floral parts—most notably the thalamus—become fleshy and contribute heavily to the main edible bulk of the fruit.

152) Explanation: Statement III is fundamentally incorrect. The functional megaspore undergoes three consecutive mitotic (not meiotic) divisions to reach the 8-nucleate stage. Statements I, II, and IV are perfectly accurate descriptions of monosporic development.

153) Explanation: Statement II is incorrect. Castor and coconut are examples of albuminous seeds, meaning they retain a significant amount of endosperm at maturity because it is not completely consumed by the developing embryo. Statements I and III are true.

154) Explanation: Scutellum is the grass cotyledon (4). Coleoptile protects the plumule (3). Epiblast is the vestigial second cotyledon (2). Coleorhiza protects the radicle (1). Thus, P-4, Q-3, R-2, S-1.

155) Explanation: Assertion (A) is FALSE because NCERT states that in chasmogamous flowers, complete autogamy is "rather rare," not "virtually prevented." (It is virtually prevented only in selfincompatible or unisexual dioecious flowers). Reason (R) is purely TRUE as it perfectly explains why complete autogamy is difficult (rare) in such open flowers.

156)

In the cross Bb × Bb, the genotypes are 1/4 BB, 1/2 Bb, and 1/4 bb. Because seed shape exhibits complete dominance, both BB and Bb are round (Total 3/4 round). Starch grain size exhibits incomplete dominance; BB is large, Bb is intermediate, bb is small. The only genotype giving BOTH round seeds AND intermediate grains is 'Bb'. The probability of Bb is 1/2.

157) Myotonic dystrophy is autosomal dominant (Let's say Dd). Unaffected female is dd. Cross:

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Dd x dd -> 1/2 Dd (affected), 1/2 dd (unaffected). Probability of child being affected is 1/2. Probability of child being female is 1/2. Total probability = 1/2 × 1/2 = 1/4.

158) Map distance is additive. Since white eye is between yellow body and miniature wing, distance(yellow to miniature) = distance(yellow to white) + distance(white to miniature) = 1.3 + 37.2 = 38.5 cM (or % recombination).

159) NCERT explicitly states the mutation is a single base substitution at the sixth codon of the beta globin gene from GAG (Glutamic acid) to GUG (Valine).

160) Alpha thalassemia is controlled by two genes (HBA1 and HBA2) on chromosome 16 (involving 4 alleles). Thalassemia in general is a quantitative defect (too few globin molecules synthesized), distinguishing it from sickle cell anemia (qualitative).

161) Genotypes: I^A i × I^B i. Probability of an O blood group child (ii) is 1/4. Probability of two consecutive O group children is (1/4) × (1/4) = 1/16.

162) Morgan coined the term linkage to describe physical association of genes and recombination to describe the generation of non-parental gene combinations.

163) In birds (domestic fowl), sex determination is ZZ/ ZW type, where the female is heterogametic (ZW) and male is homogametic (ZZ).

164) Klinefelter's (XXY) results in an overall masculine development, however, feminine development (development of breast, i.e., Gynaecomastia) is also expressed.

165) Woman's father was color-blind (X^c Y), so she inherited X^c. Mother was normal (assuming homozygous XX for this calculation since disease is rare, though could be carrier, but "normal" usually defaults to XX in simple problems unless stated otherwise). Woman is carrier X^C X^c. Man is color-blind (X^c Y). Female progeny will be 50% X^C X^c (carrier) and 50% X^c X^c (color-blind).

166) The accumulated phenylalanine is converted into phenylpyruvic acid and other derivatives, accumulating in the brain causing mental retardation.

167) Henking (1891) observed a specific nuclear structure and called it the "X body", which was later found to be the X chromosome.

168) Probability of Aa = 2/4. Probability of BB = 1/4. Probability of AaBB = (2/4) * (1/4) = 2/16.

169) If normal parents have affected offspring, the trait must be recessive (eliminates C). Since the daughter is affected, the father would have to be affected if it were X-linked

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recessive (since she needs an affected X from him). Since he is normal, it cannot be X-linked recessive (eliminates B). It must be autosomal recessive.

170) The mutant mRNA codon is GUG (codes for Valine), replacing the normal GAG (codes for Glutamic acid).

171) Both genes and chromosomes occur in pairs, and both segregate at the time of gamete formation such that only one of each pair is transmitted to a gamete (Sutton and Boveri).

172) Females are XX (have two sex chromosomes) and males are XO (have only one sex chromosome). Therefore, males have one chromosome less than females.

173) This is a symptom of Turner's syndrome, not Down's syndrome. Down's syndrome features include short statured with small round head, furrowed tongue, partially open mouth, and broad palm with characteristic palm crease.

174) For a female to be haemophilic (X^h X^h), she must receive one X^h from her mother (who must be a carrier or haemophilic) and one X^h from her father (who must be haemophilic). Since haemophilia can be severe, affected males historically had lower reproductive fitness, making this combination rare.

175) Distance A-B = 5, B-C = 15, A-C = 20. Since 5 + 15 = 20 (A-B + B-C = A-C), B is exactly between A and C. Order is A-B-C.

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176) Both A and R are true, and Reason correctly explains why the father determines the sex: he is heterogametic (XY) and contributes either X or Y, whereas the mother only contributes X.

177) All three statements are directly sourced from the NCERT textbook regarding the history of Mendel's work.

178) All three statements are true. Deletions/insertions cause frameshifts; cancer cells frequently exhibit chromosomal aberrations; and colour blindness prevalence rates are 8% for males and 0.4% for females (as stated in NCERT).

179) Solid/shaded symbols represent AFFECTED individuals, not unaffected. The other statements correctly describe standard pedigree symbols.

180) Down's is Trisomy 21. Turner's is 44+XO. Klinefelter's is 44+XXY. Beta-thalassemia is caused by mutation on chromosome 11.

Watermarkly

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